

Azure Firewall

Introduction

In today's cloud-based environments, implementing robust network security controls is essential to safeguarding resources and managing traffic flow. This lab focuses on deploying and configuring Azure Firewall to control both inbound and outbound traffic in a virtual network. By following a structured set of exercises, you will gain practical experience in creating a secure network infrastructure that includes subnets, virtual machines, custom routing, application rules, and network rules. The goal is to ensure that only authorized traffic is allowed through, thereby enforcing a secure and manageable network perimeter in Microsoft Azure.

Lab Scenario

This is a scenario where one is tasked with deploying and configuring Azure Firewall as part of a broader effort to secure network traffic within an organization's cloud infrastructure. The objective is to implement a virtual network with multiple subnets, deploy virtual machines within those subnets, and enforce outbound traffic control using firewall rules. The scenario involves creating a custom route that ensures all traffic from the workload subnet is routed through the firewall, setting up application rules to restrict web access to specific domains (such as www.bing.com), and configuring network rules to allow DNS lookups through designated external servers. All resources for this lab are to be deployed in the East US region, and the setup serves as a practical example of how Azure Firewall can be used to enhance network security and traffic visibility in a cloud environment.

Lab Objective

The objective of this lab is to demonstrate how one can deploy and configure Azure Firewall to manage and secure outbound network traffic in a virtual network environment. This includes provisioning a virtual network with separate subnets for workload and jump host virtual machines, deploying the Azure Firewall, creating custom routes to direct traffic through the firewall, and configuring both application and network rules to control access to specific resources. Additionally, the lab involves setting up DNS servers and testing the firewall's effectiveness in allowing or denying traffic based on defined rules.

Tasks

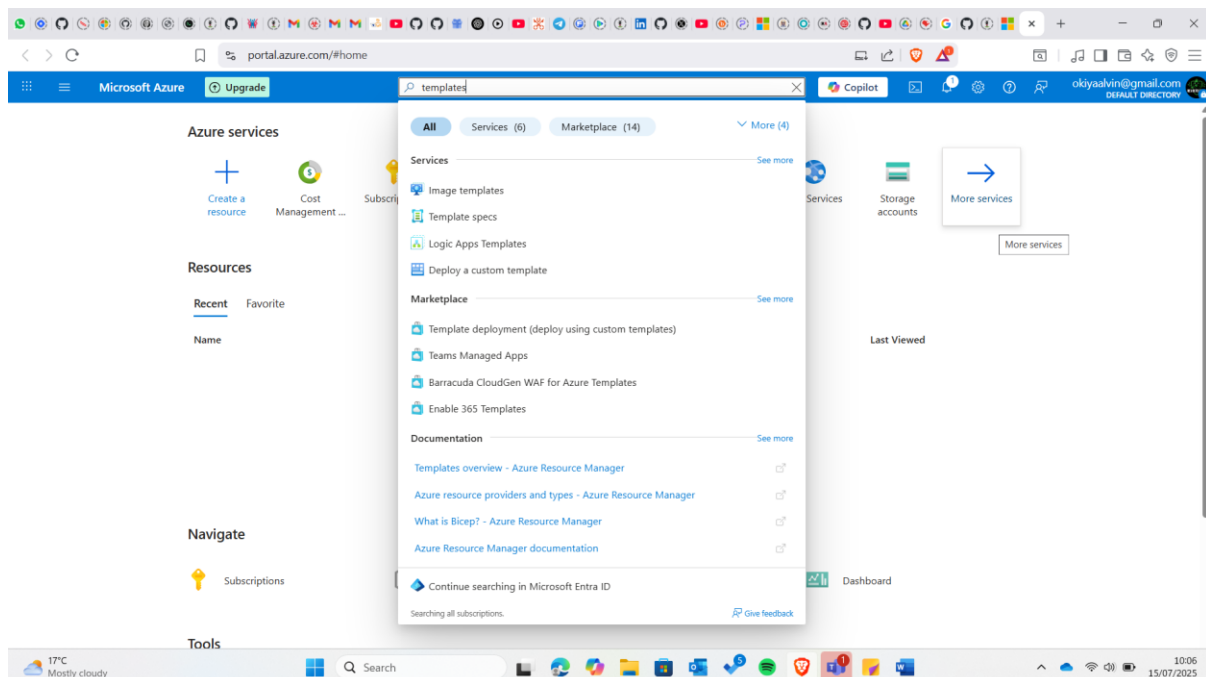
By the end of this lab, I will have done the tasks below:

1. **Used a template to deploy the lab environment** – Deployed a virtual network and virtual machines using an ARM template.

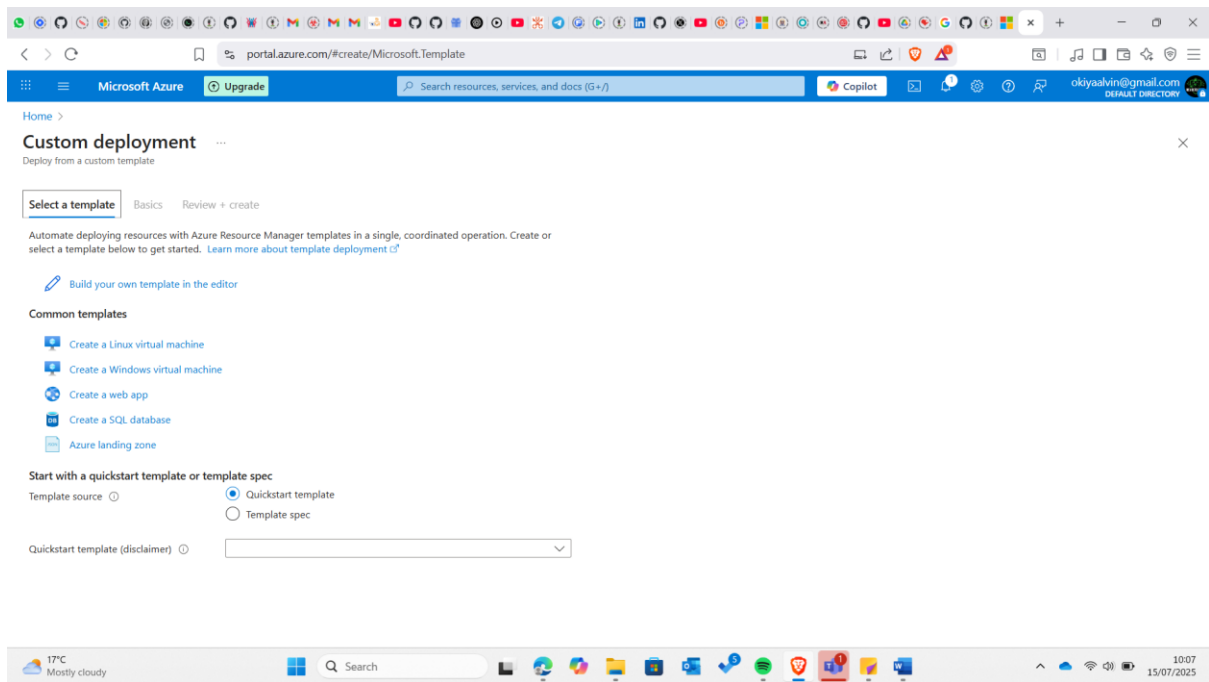
2. **Deployed an Azure Firewall** – Set up the Azure Firewall within the virtual network.
3. **Created a default route** – Configured a custom route to ensure all outbound traffic from the workload subnet is routed through the firewall.
4. **Configured an application rule** – Allowed outbound access specifically to www.bing.com using a firewall application rule.
5. **Configured a network rule** – Enabled outbound DNS traffic by allowing communication with specified external DNS servers.
6. **Configured DNS servers** – Set custom DNS server addresses for the virtual machine network interface.
7. **Tested the firewall** – Verified that only allowed traffic passed through while unauthorized traffic was blocked, confirming the firewall's functionality.

Task 1: Use a template to deploy the lab environment.

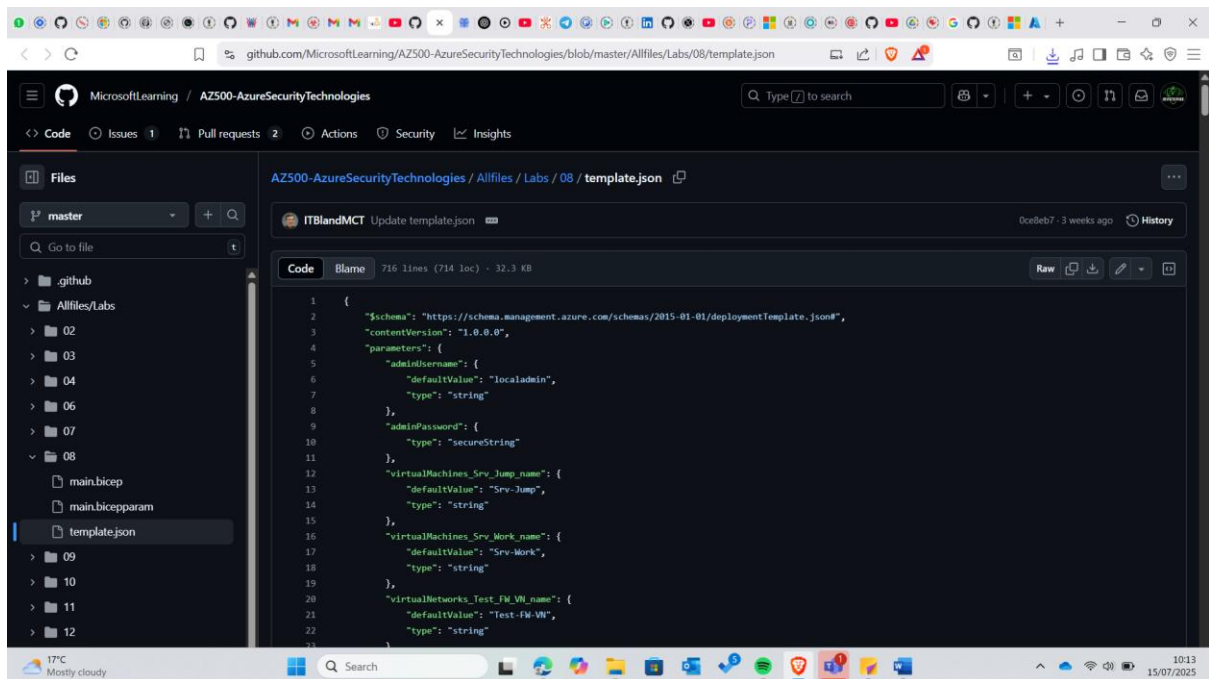
I logged into my azure portal and deployed a custom template



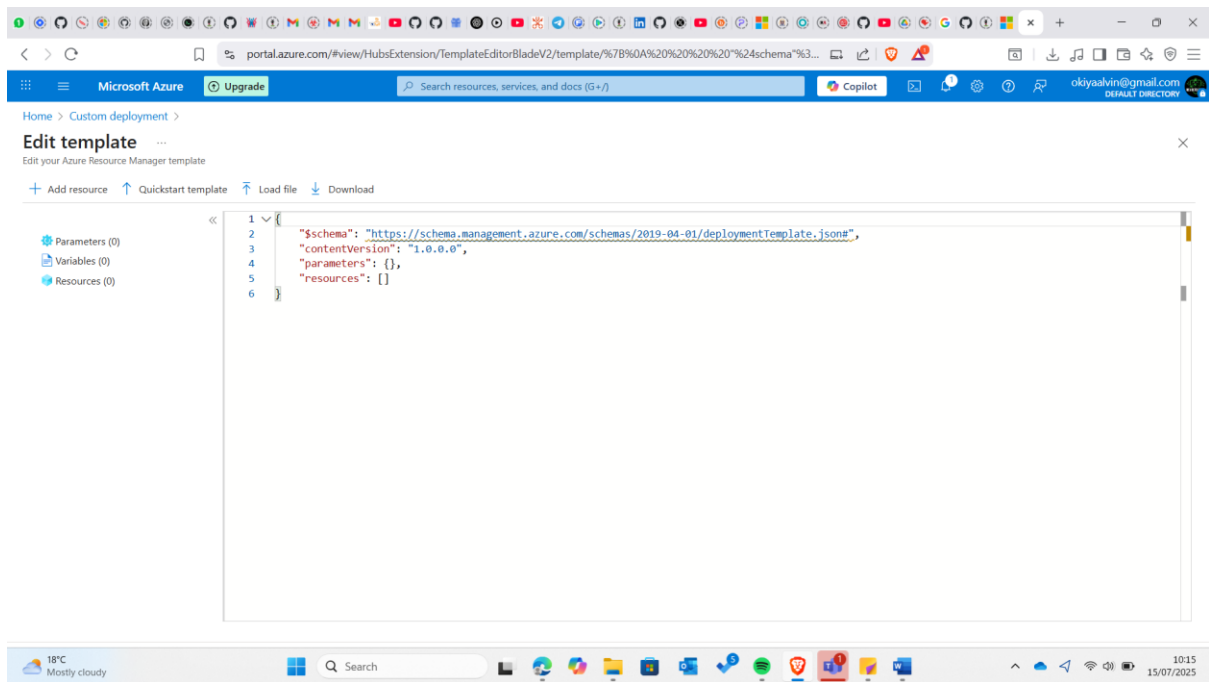
To do this I went to the search bar and searched for templates and Deploy a custom template was the 4th option. After clicking on it redirected me to the interface below:



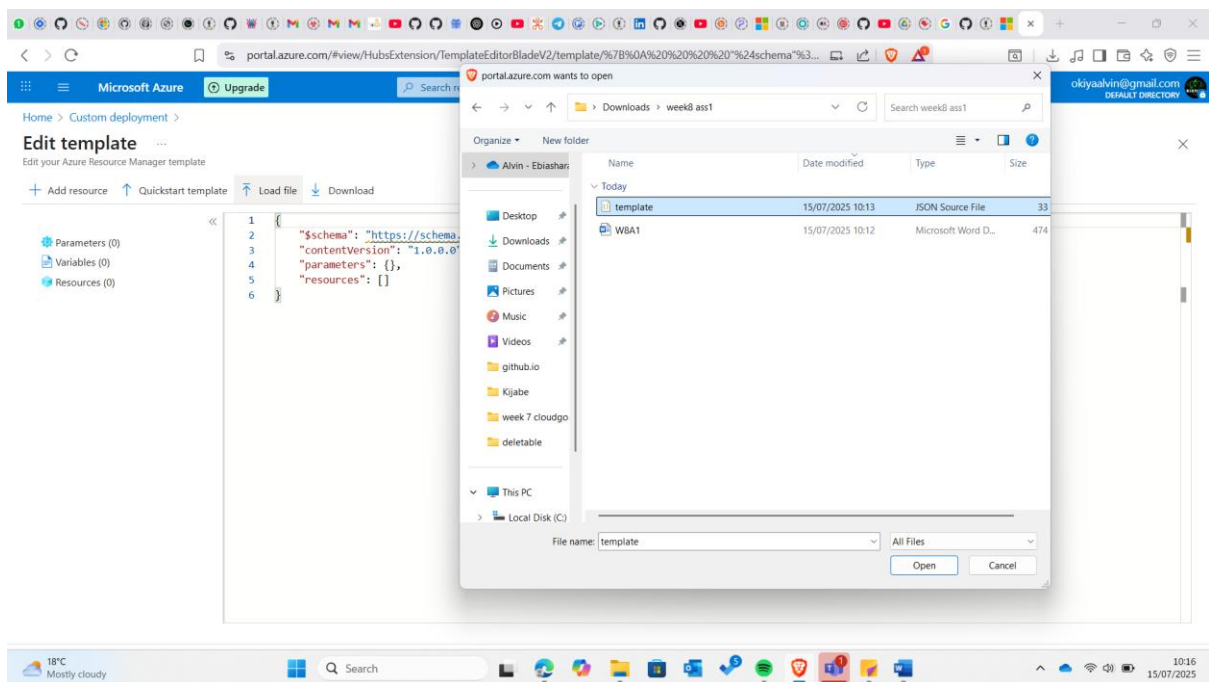
I then downloaded the json template from github:



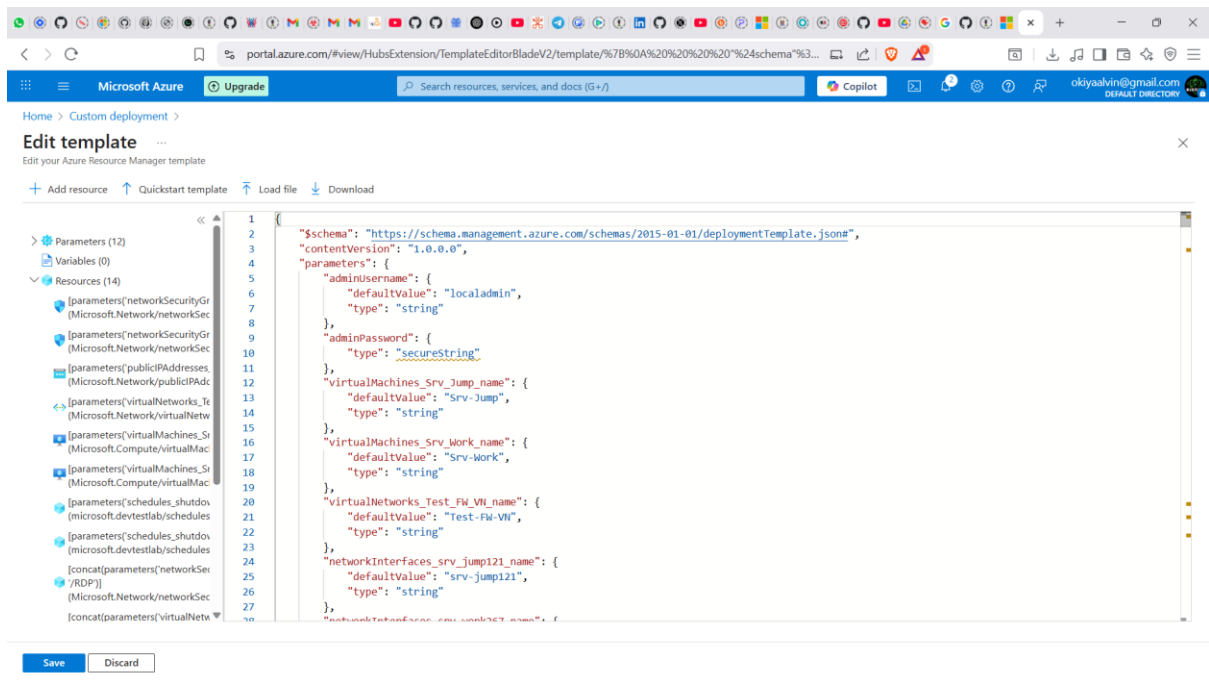
Then uploaded it on azure:



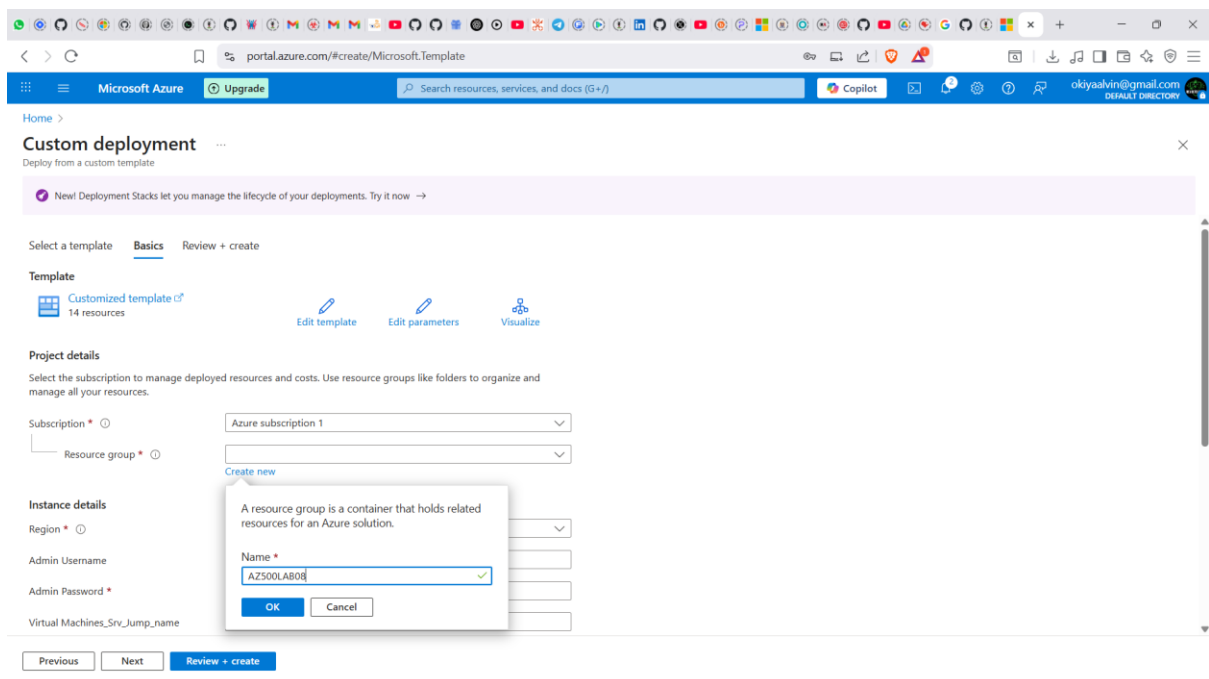
On this page I clicked on load file, which opened folders, I navigated to the location where I downloaded the json template and uploaded it:



Portalview after uploading:



Then clicked on save, Then deployed it against a new resource group:



Entered an admin password, then clicked review and create, this verifies if all details needed have been provided, before deployment begins

The screenshot shows the 'Custom deployment' page in the Azure portal. The 'Project details' section is active, where users select a subscription and resource group. Below this, the 'Instance details' section contains various configuration fields, each with a dropdown menu and a green checkmark indicating they are filled. At the bottom, there are three buttons: 'Previous', 'Next', and 'Review + create'.

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Custom deployment

Deploy from a custom template

New! Deployment Stacks let you manage the lifecycle of your deployments. Try it now →

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * (New) AZ500LAB08 [Create new](#)

Instance details

Region * East US

Admin Username localadmin

Admin Password *

Virtual Machines_Srv_Jump_name Srv-Jump

Virtual Machines_Srv_Work_name Srv-Work

Virtual Networks_Test_FW_VN_name Test-FW-VN

Network Interfaces_srv_jump121_name srv-jump121

Network Interfaces_srv_work267_name srv-work267

[Previous](#) [Next](#) [Review + create](#)

Then It takes one to this page, so I clicked on create, to start the deployment:

The screenshot shows the 'Review + create' section of the 'Custom deployment' page. It includes a summary of the deployment, terms and conditions, and a 'Create' button at the bottom. The 'Summary' section shows a 'Customized template' with '14 resources'. The 'Terms' section contains legal disclaimers and a 'Create' button.

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Custom deployment

Deploy from a custom template

Select a template Basics **Review + create**

Summary

Customized template

14 resources

Terms

[Azure Marketplace Terms](#) | [Azure Marketplace](#)

By clicking "Create," I (a) agree to the applicable legal terms associated with the offering; (b) authorize Microsoft to charge or bill my current payment method for the fees associated with the offering(s), including applicable taxes, with the same billing frequency as my Azure subscription, until I discontinue use of the offering(s); and (c) agree that, if the deployment involves 3rd party offerings, Microsoft may share my contact information and other details of such deployment with the publisher of that offering.

Microsoft assumes no responsibility for any actions performed by third-party templates and does not provide rights for third-party products or services. See the [Azure Marketplace Terms](#) for additional terms.

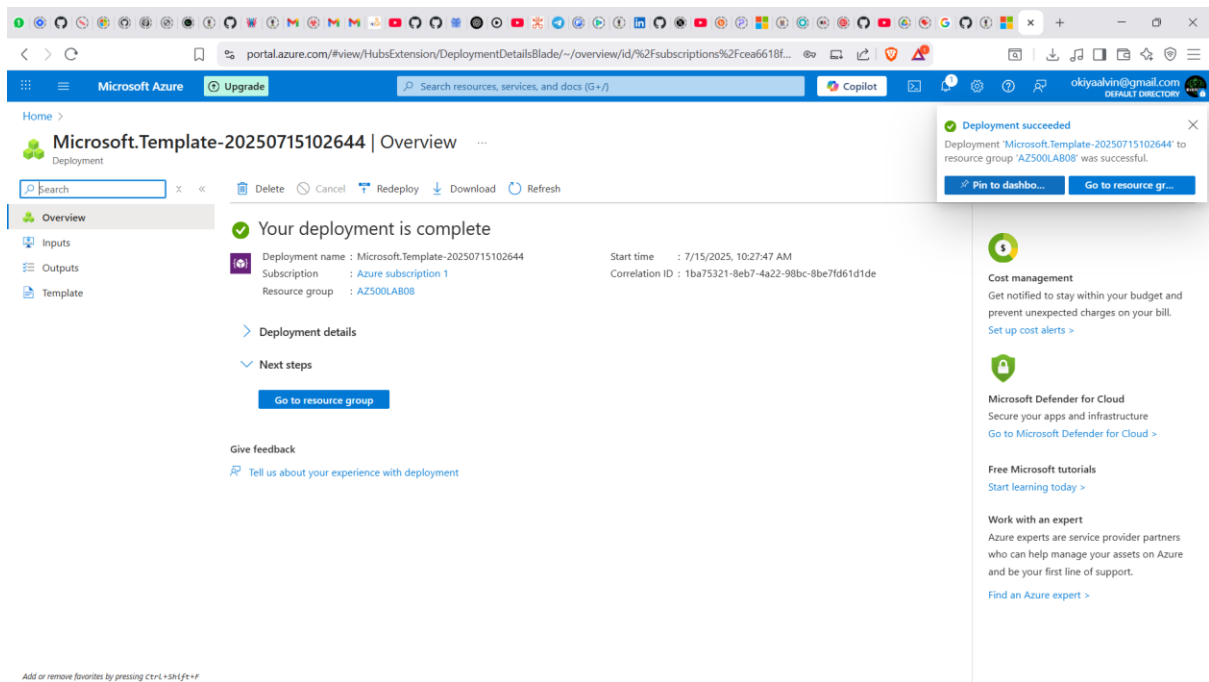
Deploying this template will create one or more Azure resources or Marketplace offerings. You acknowledge that you are responsible for reviewing the applicable pricing and legal terms associated with all resources and offerings deployed as part of this template. Prices and associated legal terms for any Marketplace offerings can be found in the [Azure Marketplace](#); both are subject to change at any time prior to deployment.

Neither subscription credits nor monetary commitment funds may be used to purchase non-Microsoft offerings. These purchases are billed separately.

If any Microsoft products are included in a Marketplace offering (e.g. Windows Server or SQL Server), such products are licensed by Microsoft and not by any third party.

[Previous](#) [Next](#) [Create](#)

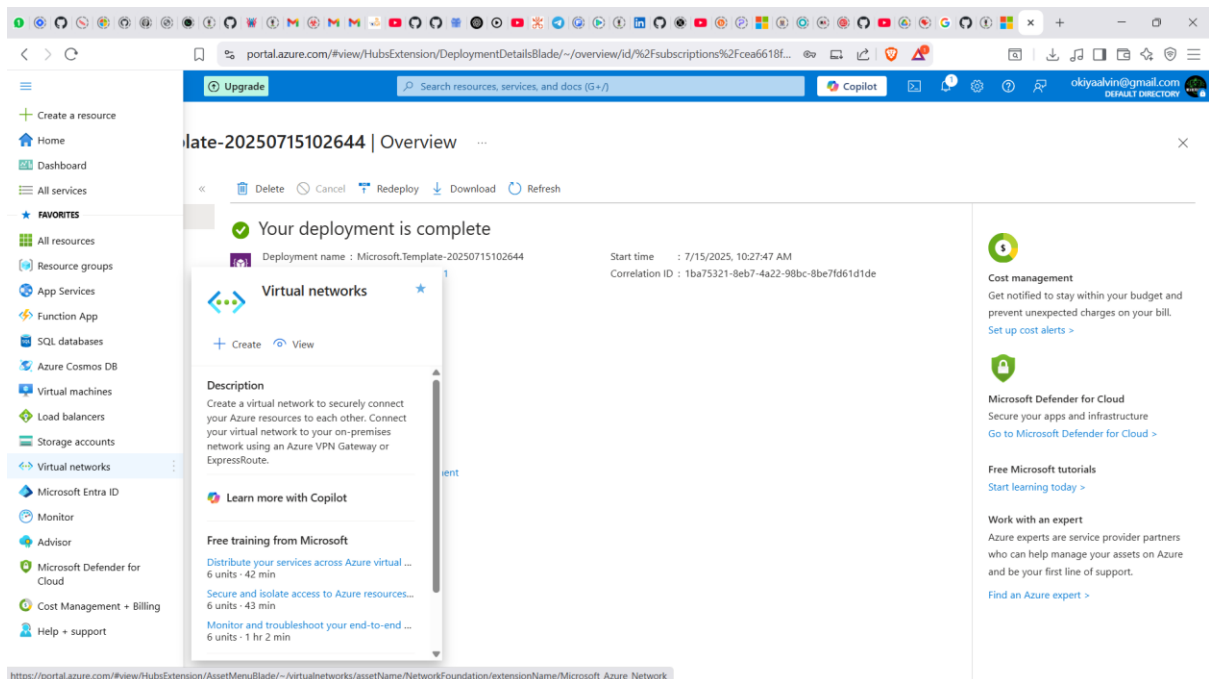
Once the deployment was complete, it looked this way:



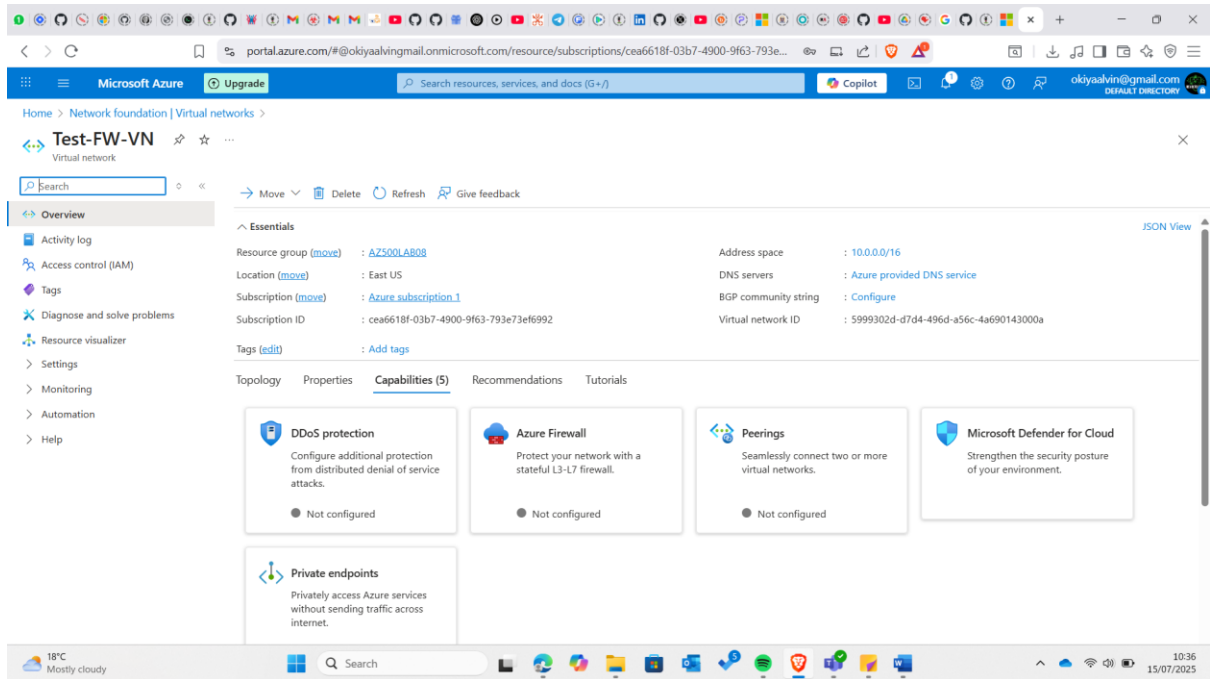
This marks a successful deployment of the lap environment.

Task 2: Deploy an Azure firewall.

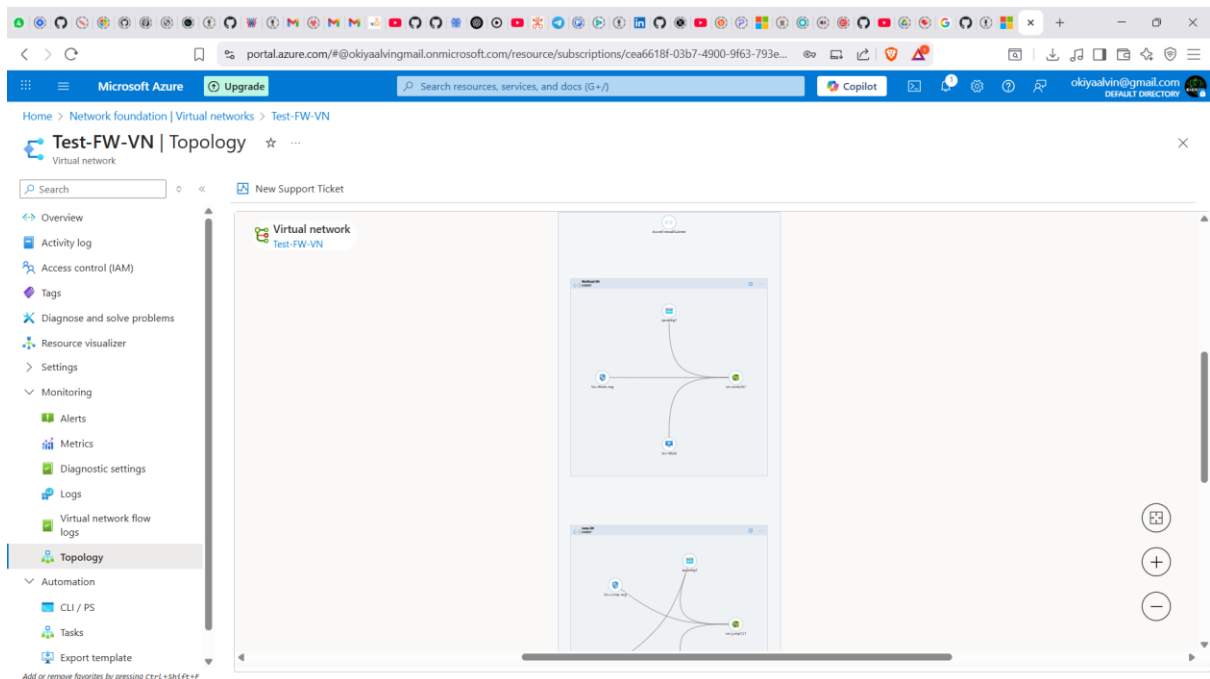
Once the deployment is complete, I navigated to the virtual Network section of the portal:



Then to the template which is called TEST-FW-VN:

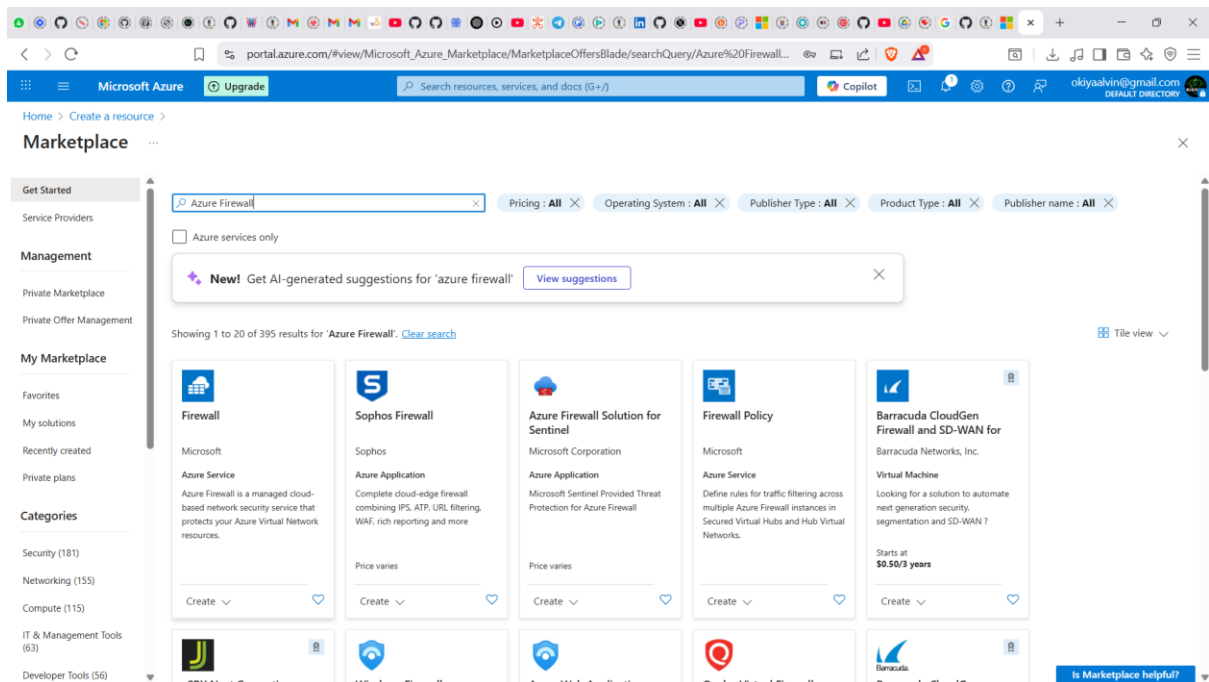


After viewing the topology,

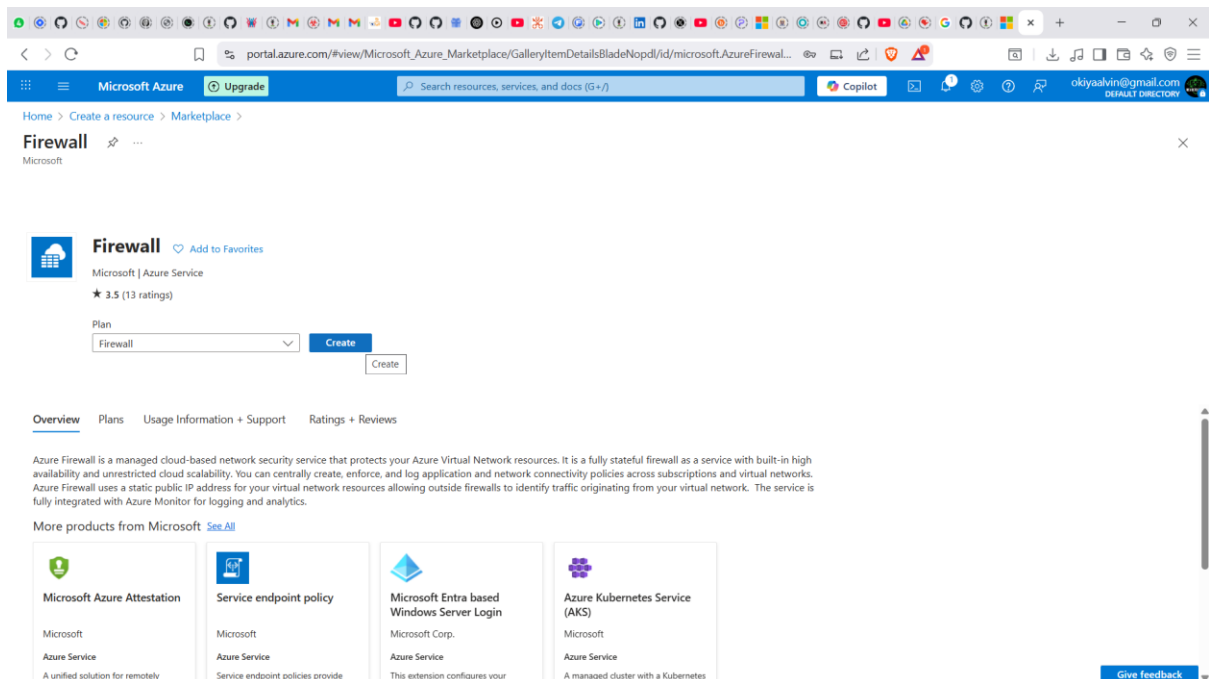


I proceeded to create an azure firewall appliance:

To this I clicked on create resource then searched for Azure Firewall:



I clicked on it, then clicked on create:



I then filled the following details:

portal.azure.com/#create/microsoft.AzureFirewall-Arm

Microsoft Azure Upgrade Search resources, services, and docs (5+)

Home > Create a resource > Marketplace > Firewall >

Create a firewall

Basics Tags Review + create

Azure Firewall is a managed cloud-based network security service that protects your Azure Virtual Network resources. It is a fully stateful firewall as a service with built-in high availability and unrestricted cloud scalability. You can centrally create, enforce, and log application and network connectivity policies across subscriptions and virtual networks. Azure Firewall uses a static public IP address for your virtual network resources allowing outside firewalls to identify traffic originating from your virtual network. The service is fully integrated with Azure Monitor for logging and analytics. [Learn more](#)

Project details

Subscription * Azure subscription 1

Resource group * AZ500LAB08 [Create new](#)

Instance details

Name * Test-FW01

Region * East US

Firewall SKU

☐ Basic

☒ Standard

[Previous](#) [Next: Tags >](#) [Download a template for automation](#)

<https://portal.azure.com/#>

Added a new public ip “TEST-FW-PIP” and disabled the NIC option:

portal.azure.com/#create/microsoft.AzureFirewall-Arm

Microsoft Azure Upgrade Search resources, services, and docs (5+)

Home > Create a resource > Marketplace > Firewall >

Create a firewall

☒ Standard

☐ Premium

Premium firewalls support additional capabilities, such as SSL termination and IDPS. Additional costs may apply. [Learn more](#)

Firewall management

☐ Use a Firewall Policy to manage this firewall

☒ Use Firewall rules (classic) to manage this firewall

Choose a virtual network

☐ Create new

☒ Use existing

Virtual network

Test-FW-VN (AZ500LAB08)

Public IP address *

(New) TEST-FW-PIP [Add new](#)

Firewall Management NIC

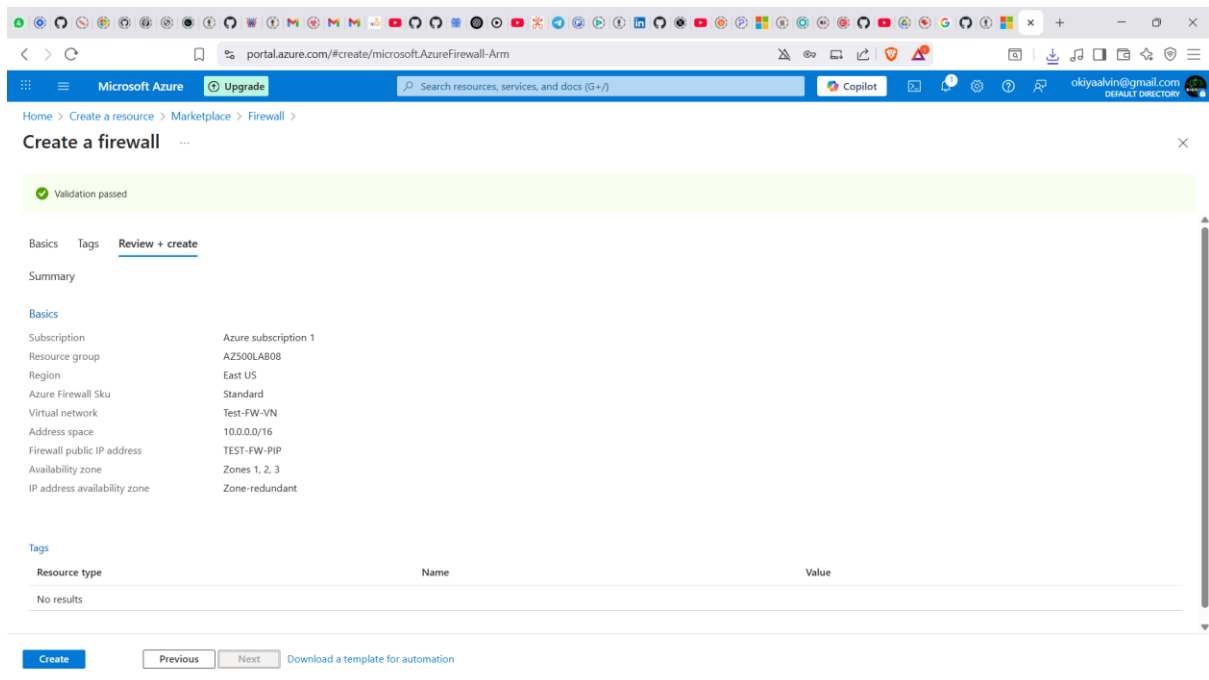
Firewall Management NIC separates Firewall management traffic from customer traffic. A dedicated subnet is required with its own associated public IP address that will be used exclusively by the Azure platform and can't be used for any other purpose.

Enable Firewall Management NIC

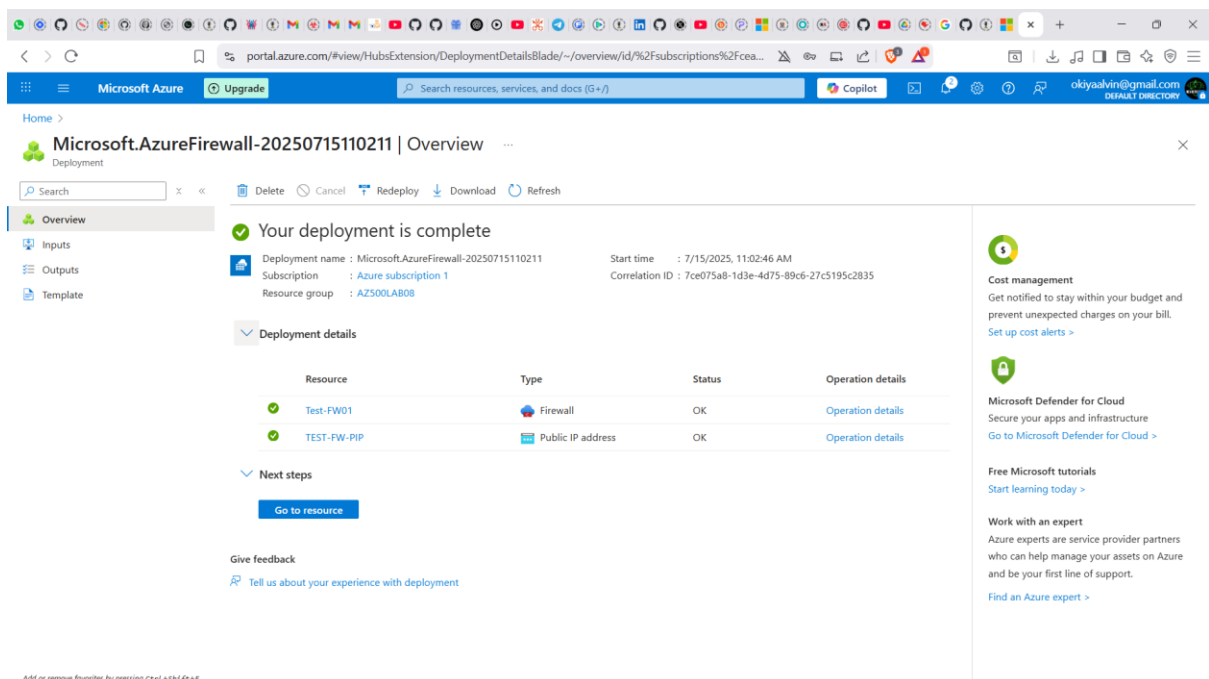
☒ **Critical Firewall features such as Forced Tunneling and Packet Capture require management NIC to be enabled.**

[Previous](#) [Next: Tags >](#) [Download a template for automation](#)

After the validation was passed I proceeded to click create:

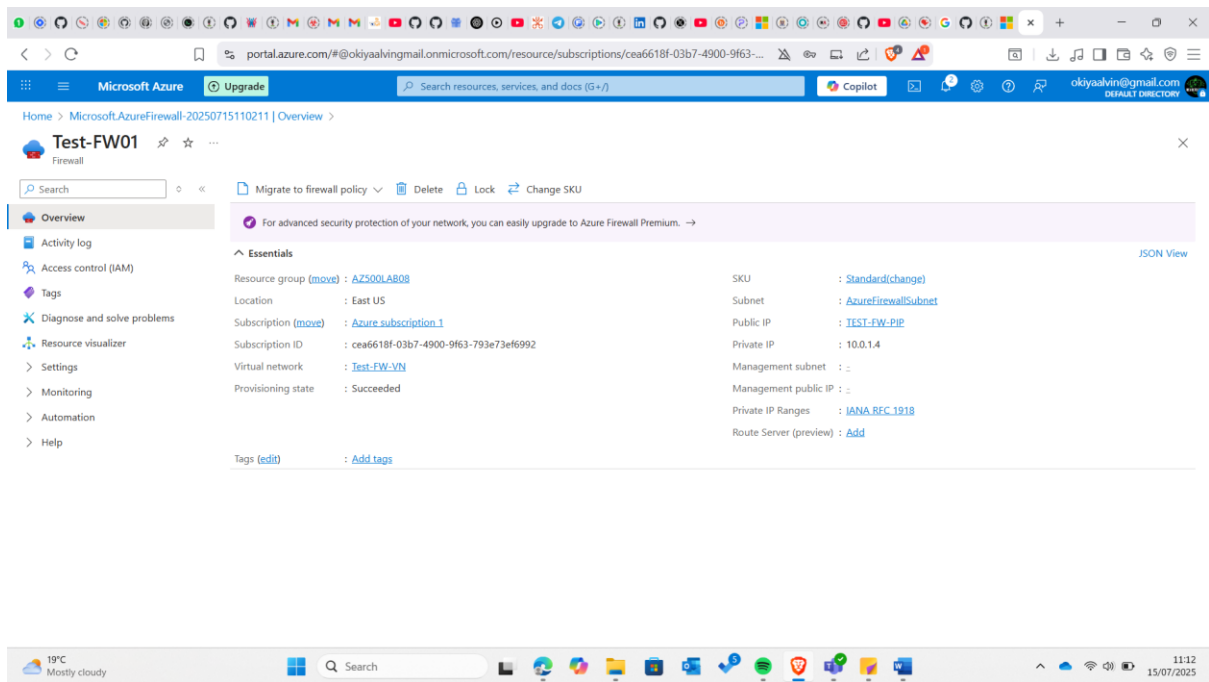


This started the deployment of the firewall: It took around 5-7 minutes:



I then clicked on “go to resource” in order to get the private ip so that later on we can deploy a User Defined Route:

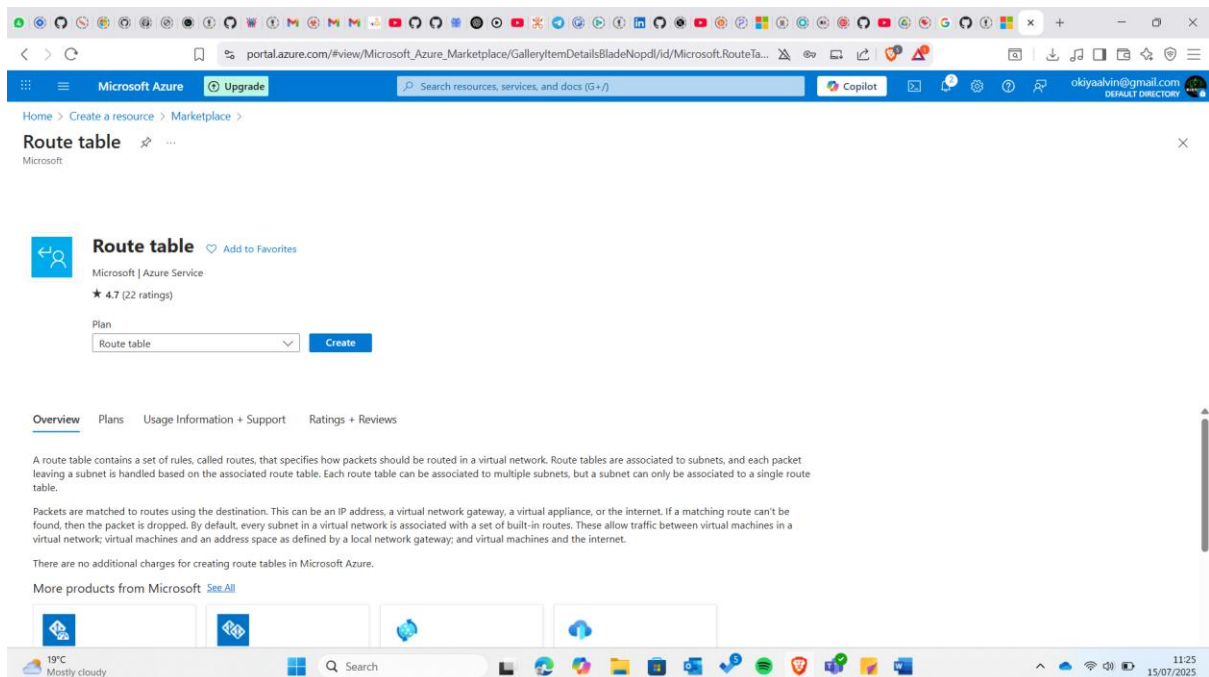
Ip : 10.0.1.4



This marks a successful Deployment of azure firewall

Task 3: Create a default route

To do this I clicked on create a resource and searched for route table and clicked on create:



Then filled the details based on the instructions, and clicked on “review and create”

The screenshot shows the 'Create Route table' page in the Microsoft Azure portal. The 'Basics' tab is selected. The page title is 'Create Route table'. Below the title, there are tabs for 'Basics', 'Tags', and 'Review + create'. The 'Project details' section instructs the user to select a subscription to manage deployed resources and costs. The 'Subscription' dropdown is set to 'Azure subscription 1' and the 'Resource group' dropdown is set to 'AZ500LAB08'. The 'Instance details' section shows the 'Region' as 'East US' and the 'Name' as 'Firewall-route'. The 'Propagate gateway routes' option is set to 'Yes'. At the bottom, there are buttons for 'Previous', 'Next', and 'Review + create'. A 'Give feedback' link is also present.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Create a resource > Marketplace > Route table >

Create Route table

Basics Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * Azure subscription 1

Resource group * AZ500LAB08 [Create new](#)

Instance details

Region * East US

Name * Firewall-route

Propagate gateway routes * ☒ Yes ☐ No

[Previous](#) [Next](#) [Review + create](#) [Give feedback](#)

It then went through the validation process upon clicking create and started deployment:

The screenshot shows the 'Create Route table' page in the Microsoft Azure portal, now in the 'Review + create' tab. The 'Basics' tab is still selected. The page title is 'Create Route table'. Below the title, there are tabs for 'Basics', 'Tags', and 'Review + create'. The 'Review + create' tab is active, showing a summary of the configuration. The 'Subscription' is 'Azure subscription 1', the 'Resource group' is 'AZ500LAB08', the 'Region' is 'East US', the 'Name' is 'Firewall-route', and the 'Propagate gateway routes' option is 'Yes'. At the bottom, there are buttons for 'Previous', 'Next', and 'Create'. A 'Give feedback' link is also present.

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Create a resource > Marketplace > Route table >

Create Route table

Basics Tags Review + create

[View automation template](#)

Basics

Subscription	Azure subscription 1
Resource group	AZ500LAB08
Region	East US
Name	Firewall-route
Propagate gateway routes	Yes

[Previous](#) [Next](#) [Create](#) [Give feedback](#)

Once the deployment was complete the interface looked as shown:

portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade~/overview/id/%2Fsubscriptions%2Fcea...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Microsoft.RouteTable-20250715112534 | Overview

Deployment

Search Delete Cancel Redeploy Download Refresh

Overview Inputs Outputs Template

✓ Your deployment is complete

Deployment name : Microsoft.RouteTable-20250715112534 Start time : 7/15/2025, 11:28:28 AM
Subscription : Azure subscription 1 Correlation ID : 2f7d14b2-b80d-407d-967e-d6bd6530974d
Resource group : AZ500LAB08

Deployment details

Resource	Type	Status	Operation details
✓ Firewall-route	Route table	OK	Operation details

Next steps

[Go to resource](#)

Give feedback

[Tell us about your experience with deployment](#)

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I then clicked on go to resource:

portal.azure.com/#@okiyaalvingmail.onmicrosoft.com/resource/subscriptions/cea6618f-03b7-4900-9f63-...

Microsoft Azure Upgrade Search resources, services, and docs (G+)

Home > Microsoft.RouteTable-20250715112534 | Overview > Firewall-route

Route table

Search Move Delete Refresh Give feedback

Overview Activity log Access control (IAM) Tags Diagnose and solve problems Resource visualizer Settings Monitoring Automation Help

Essentials

Resource group (move) : AZ500LAB08 Associations : 0 subnet associations
Location : East US
Subscription (move) : Azure subscription 1
Subscription ID : cea6618f-03b7-4900-9f63-793e73ef6992
Tags (edit) : [Add tags](#)

Routes

Search routes

Name	Address prefix	Next hop type	Next hop IP address
No results.			

Subnets

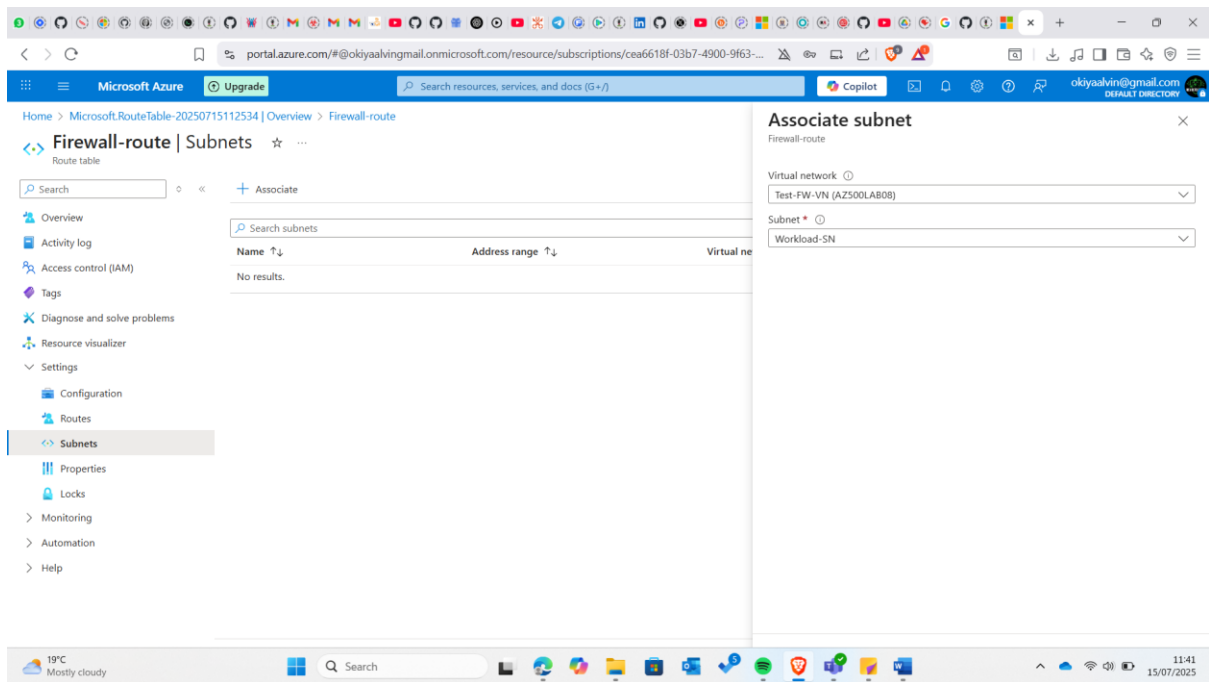
Search subnets

Name	Address range	Virtual network	Security group
No results.			

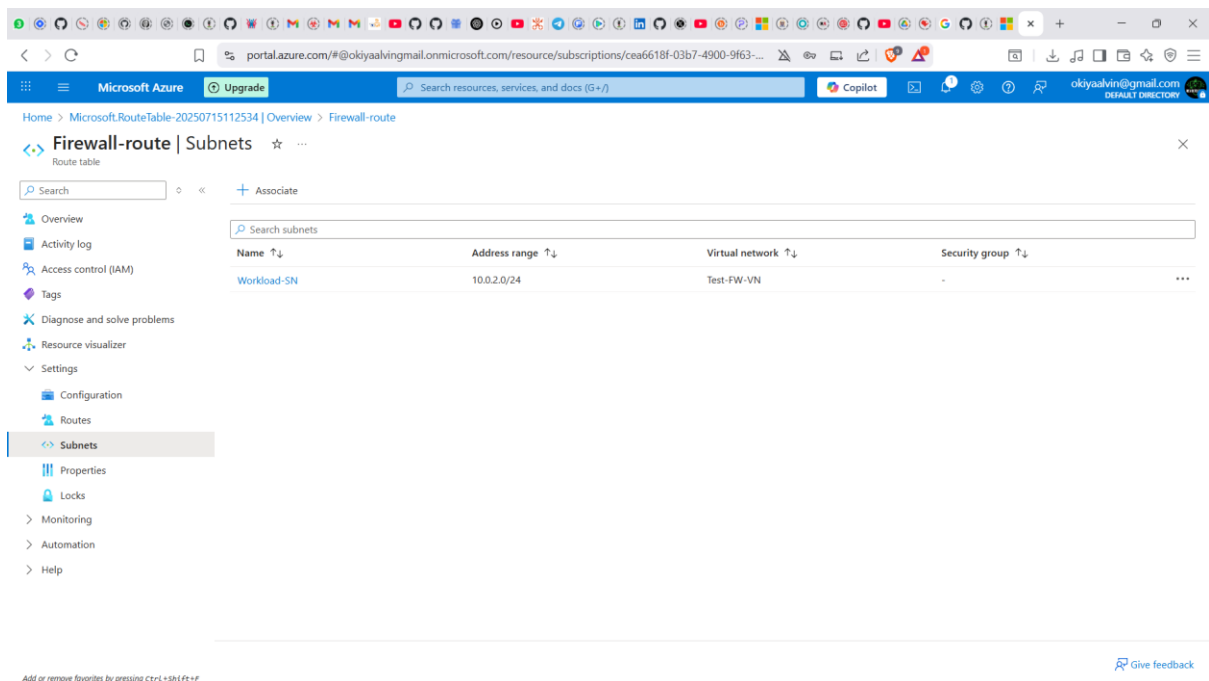
Add or remove favorites by pressing Ctrl+Shift+F

Then under settings I clicked on subnets, then associate:

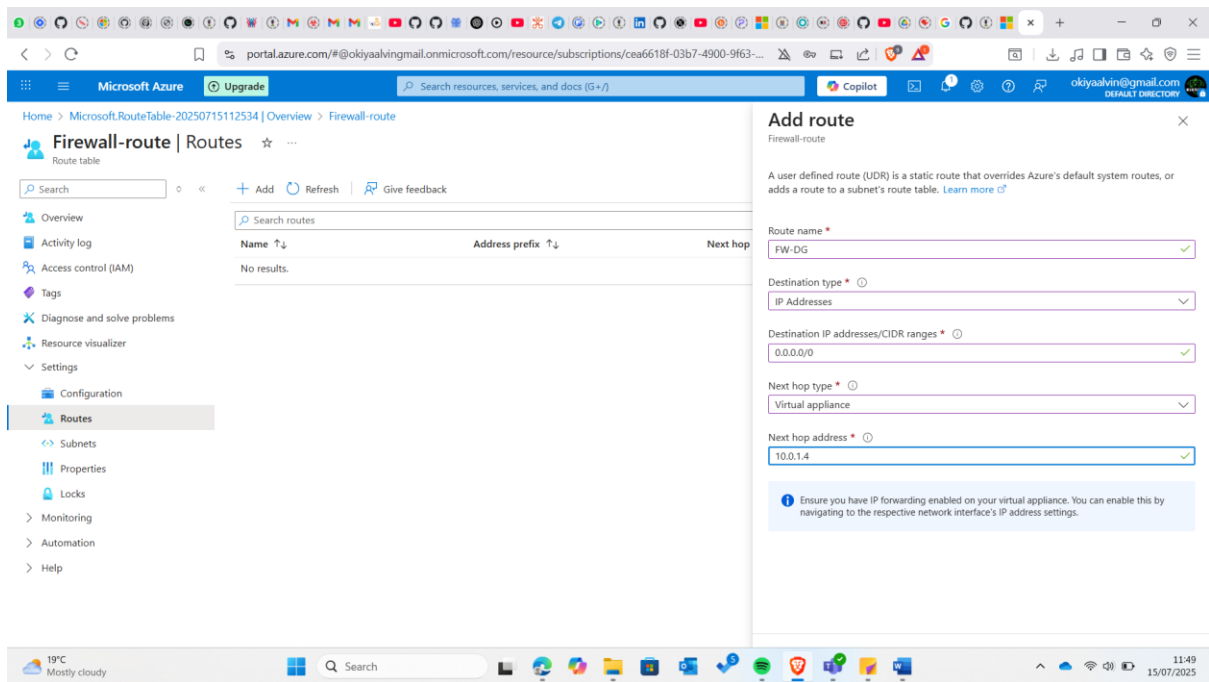
By doing this we are associating the user defined route to the subnet. Both should be in the same region.



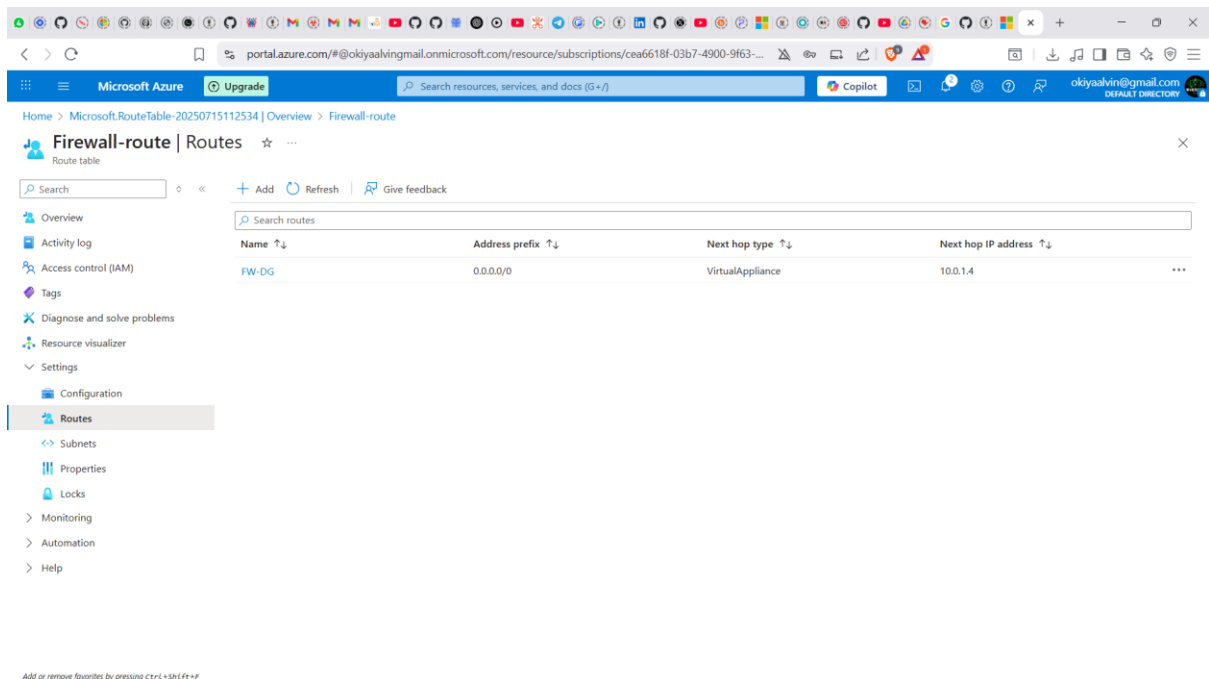
On clicking ok, the changes will be shown on the page



I then clicked on routes, under settings and clicked on add then configured the settings based on the instructions and the next hop ip based on the earlier identified ip:

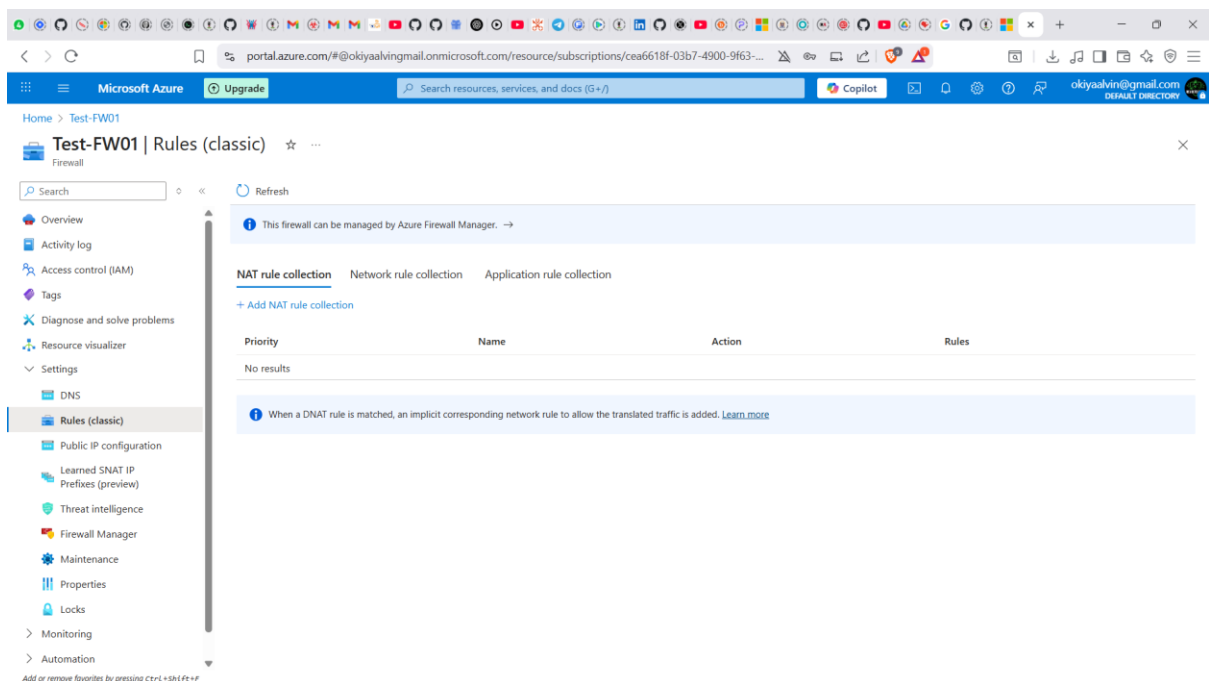
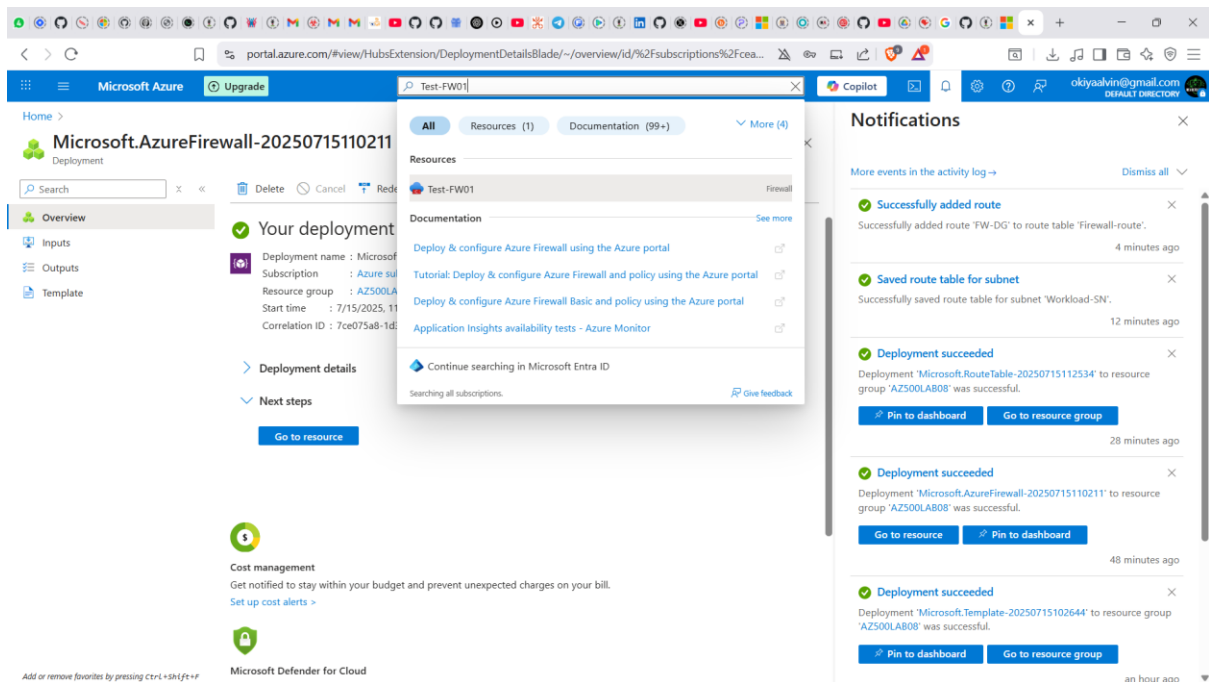


After clicking on add, the route was successfully added, marking the completion of this step:

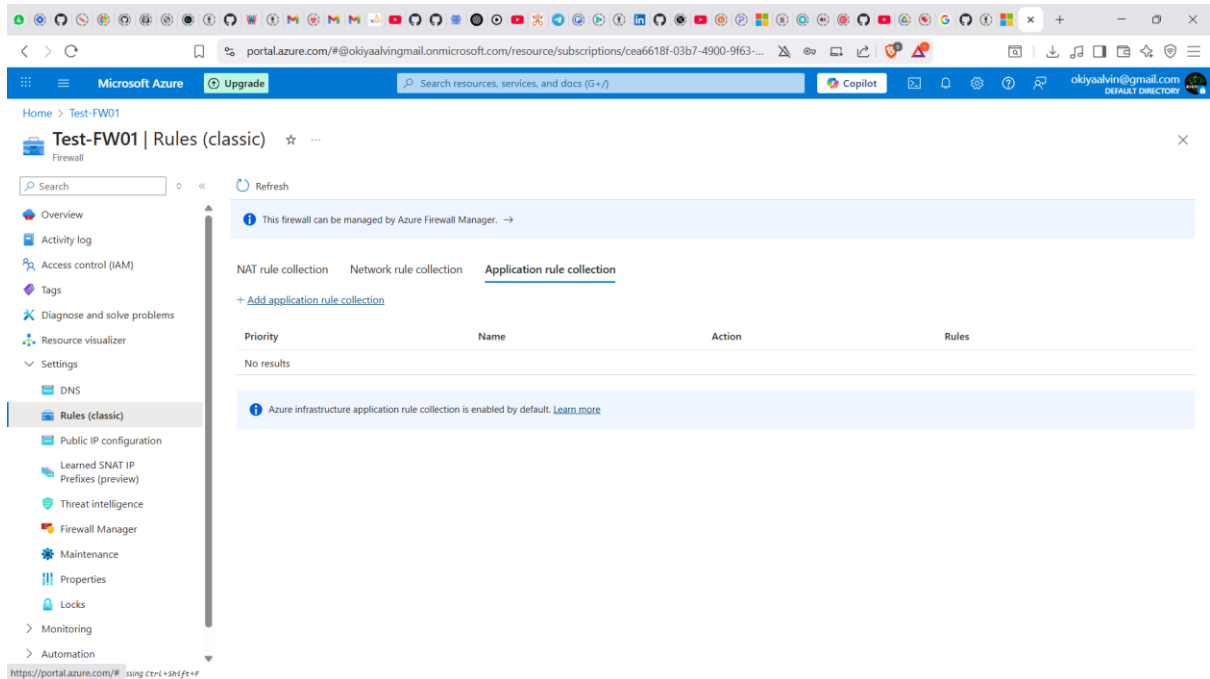


Task 4: Configure an application rule

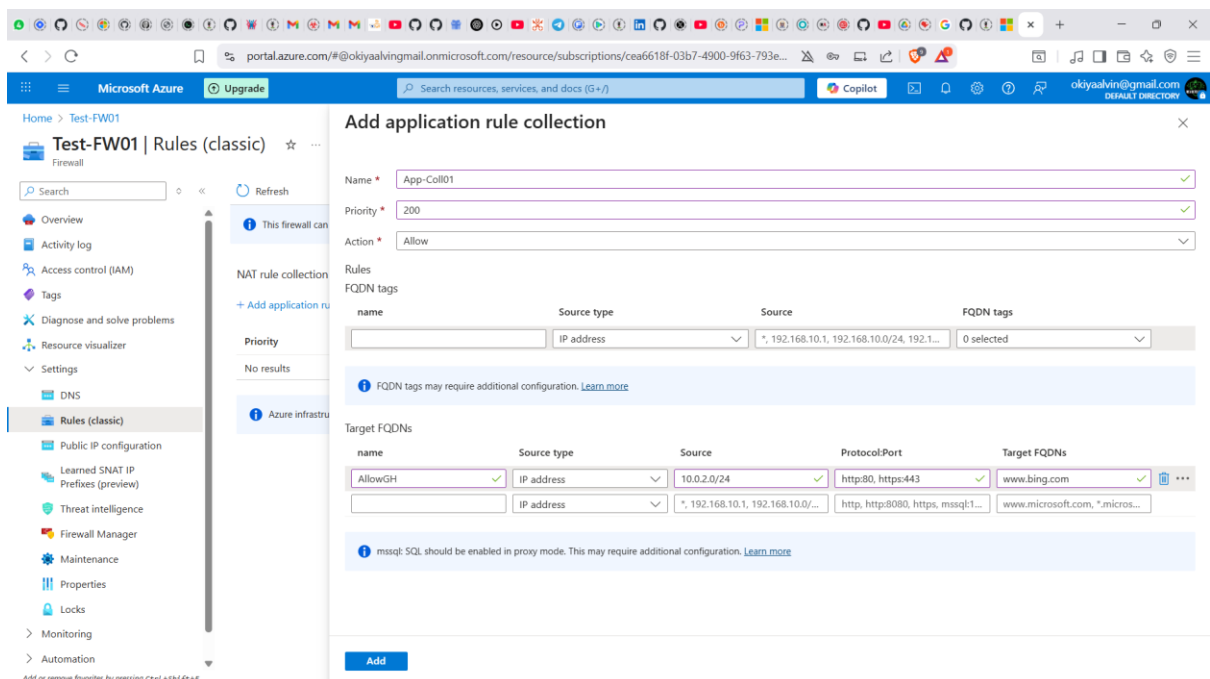
In this step, I navigated back to the **Test-FW01** firewall, and on the **Test-FW01** blade, in the **Settings** section, I clicked **Rules (classic)**



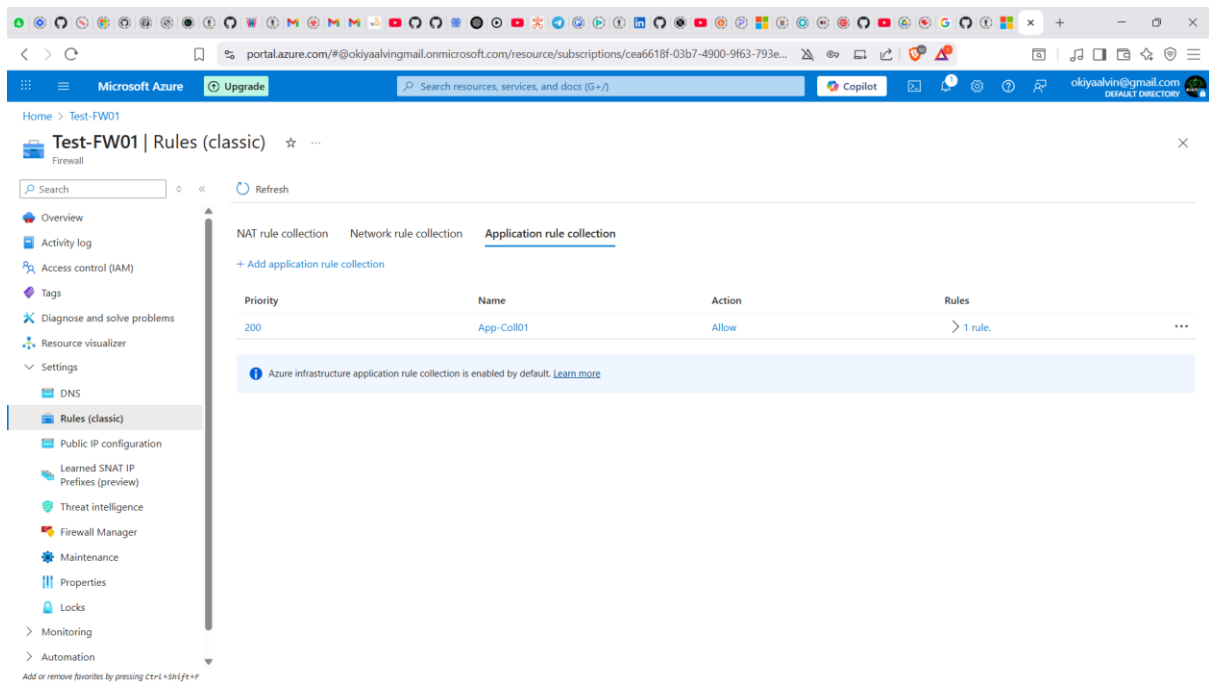
I selected the Application rule collection and clicked on Add application rule collection



I then configured the application rules as shown:



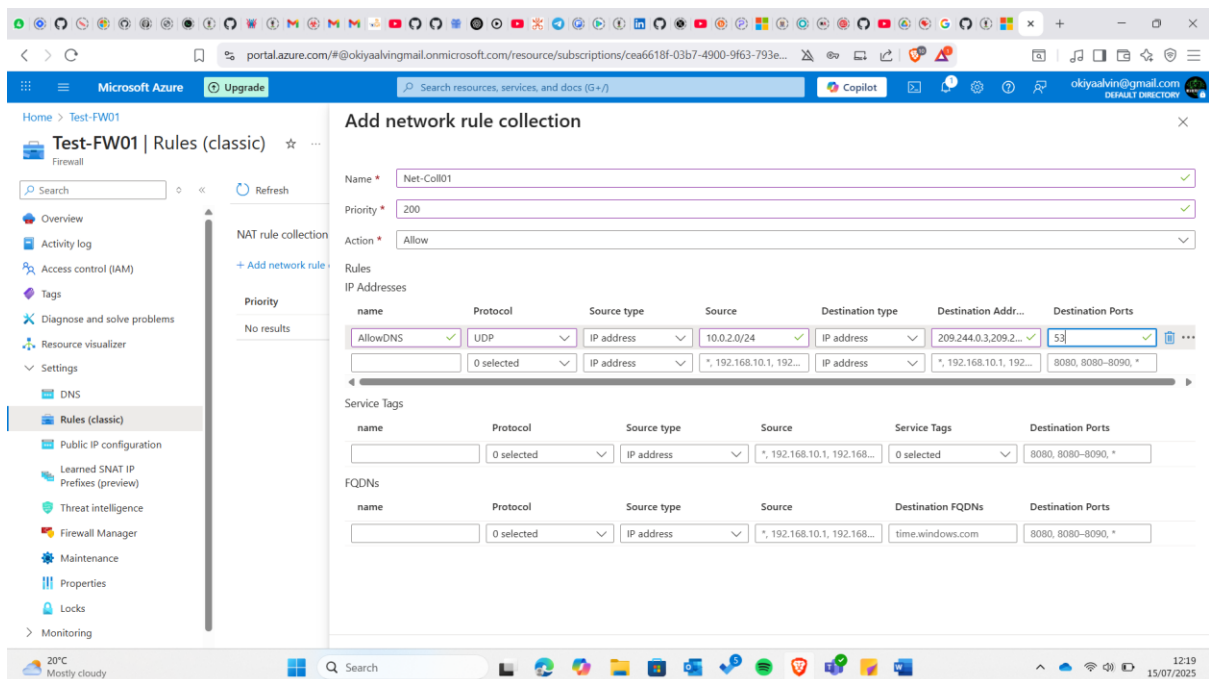
I then clicked **Add** to add the Target FQDNs-based application rule.



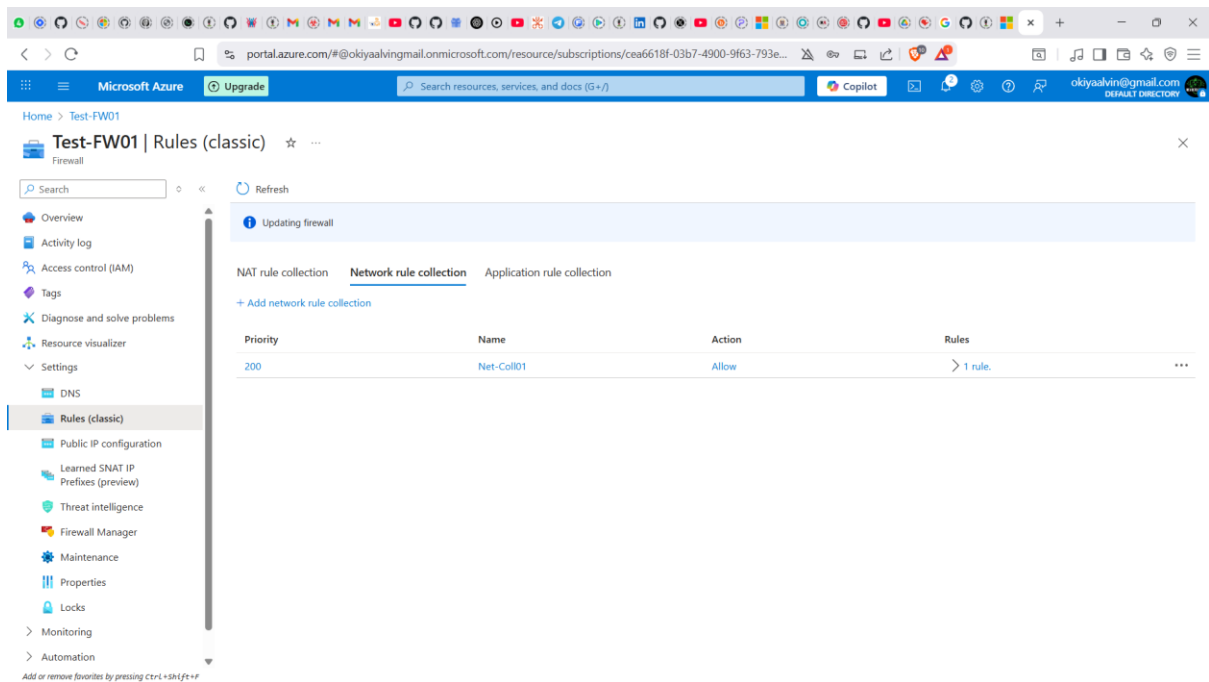
Task 5: Configure a network rule

In this task I navigated back to the **Test-FW01 | Rules (classic)** blade then clicked the **Network rule collection** tab and then clicked **+ Add network rule collection**.

On the **Add network rule collection** blade and the **Add network rule collection** blade, I specified the settings according to the provided instructions:



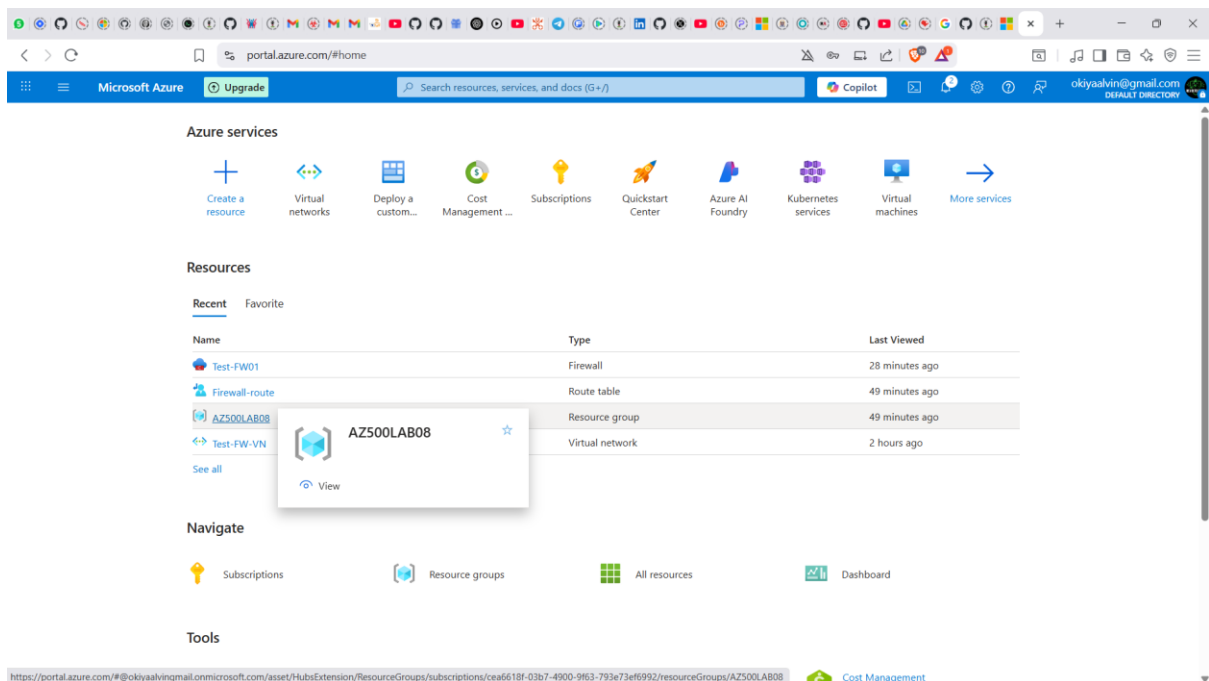
And just like before, I clicked on add and the rules were created:

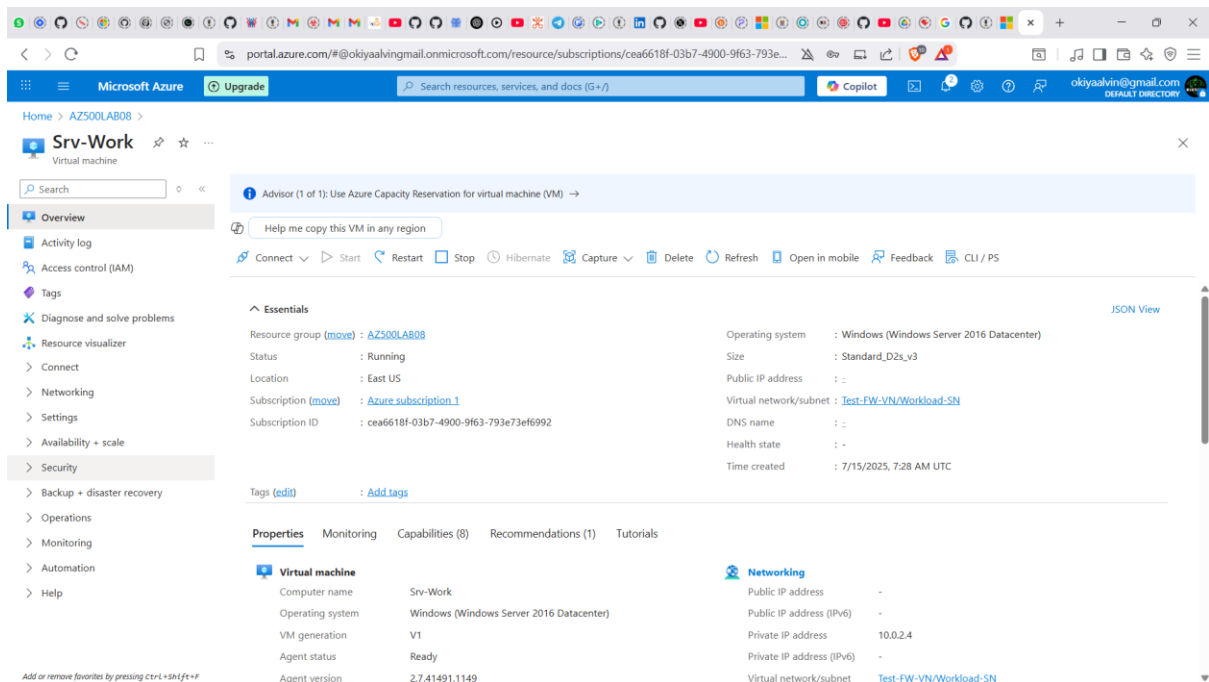


Task 6: Configure the virtual machine DNS servers

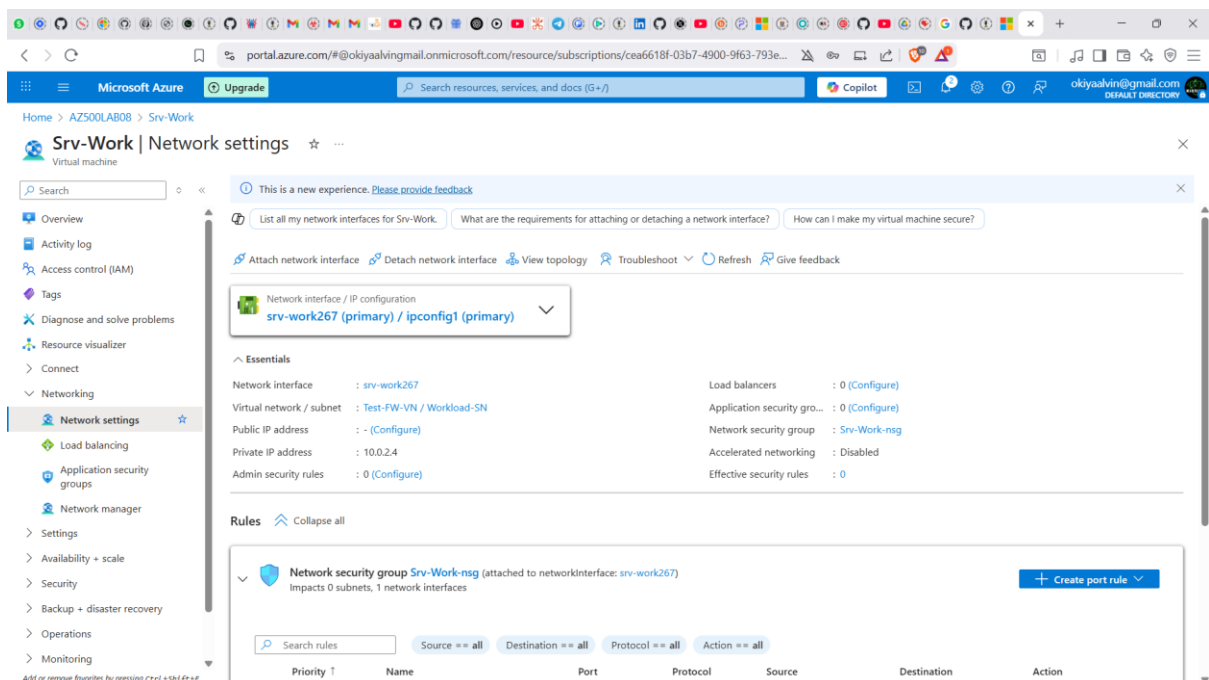
In this task, I was to configure the primary and secondary DNS addresses for the virtual machine. This is not a firewall requirement.

In the azure portal, I navigated back to the **AZ500LAB08** resource group, and on the resource blade I clicked the the **Srv-Work** virtual machine, then clicked **Networking**.

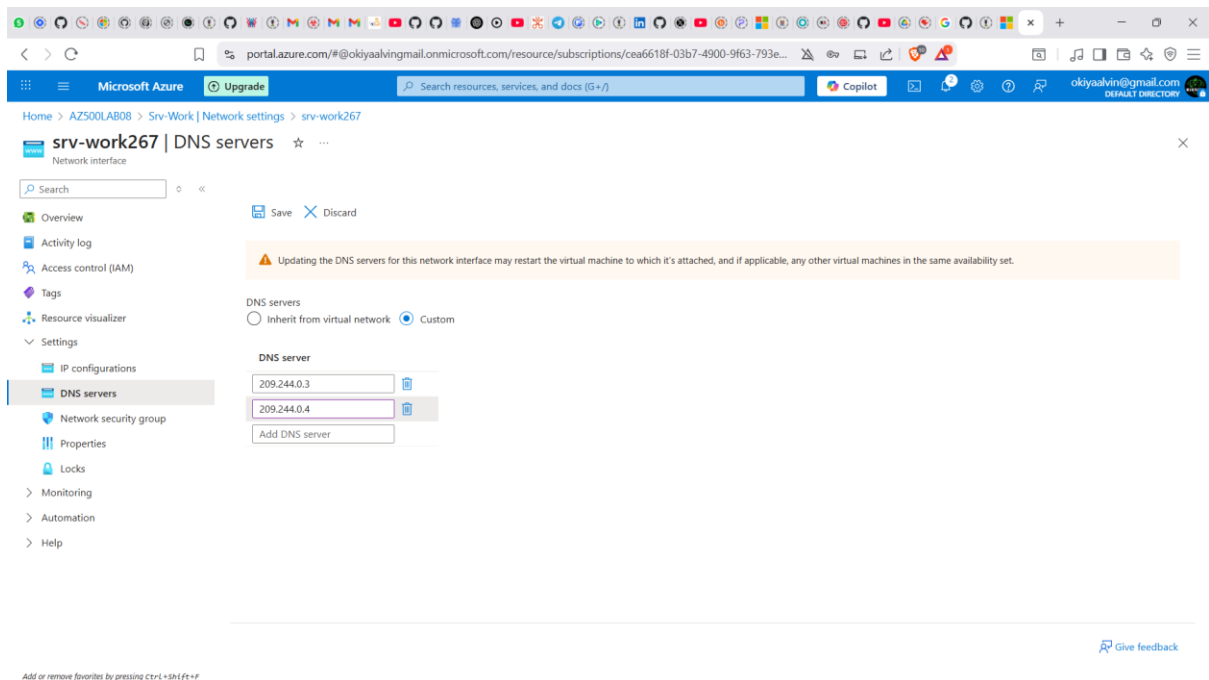




Under networking I clicked on Network settings then on the **Srv-Work | Networking Settings** blade, I clicked the link next to the **Network interface** entry:



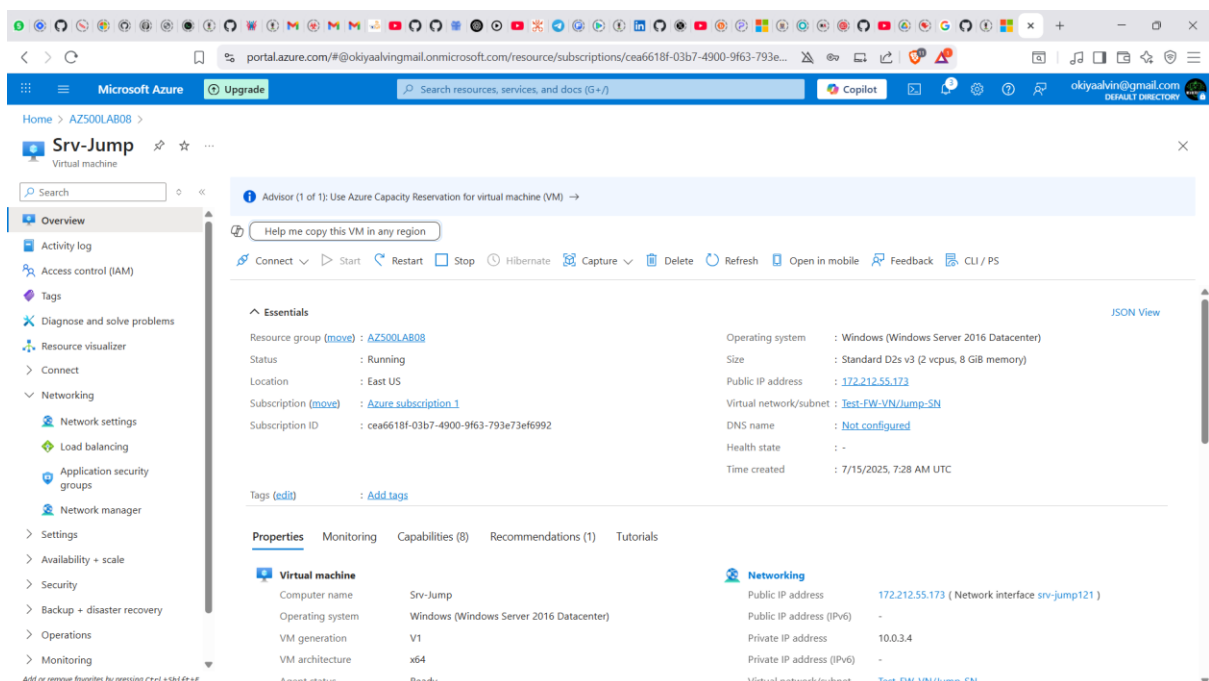
Next step, On the network interface blade, in the **Settings** section, I clicked **DNS servers**, selected the **Custom** option, added the two DNS servers referenced in the network rule: **209.244.0.3** and **209.244.0.4**, and clicked **Save** to save the change:



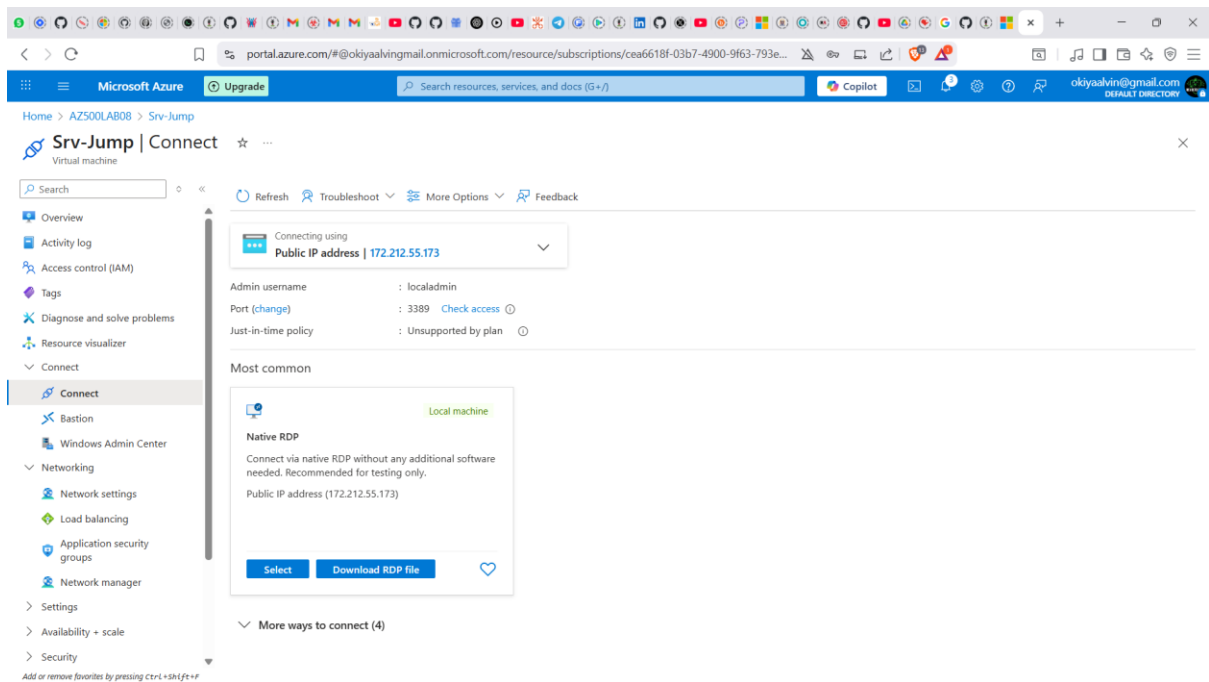
I then returned to the **Srv-Work** virtual machine page waiting for the update to complete.

Task 7: Test the firewall

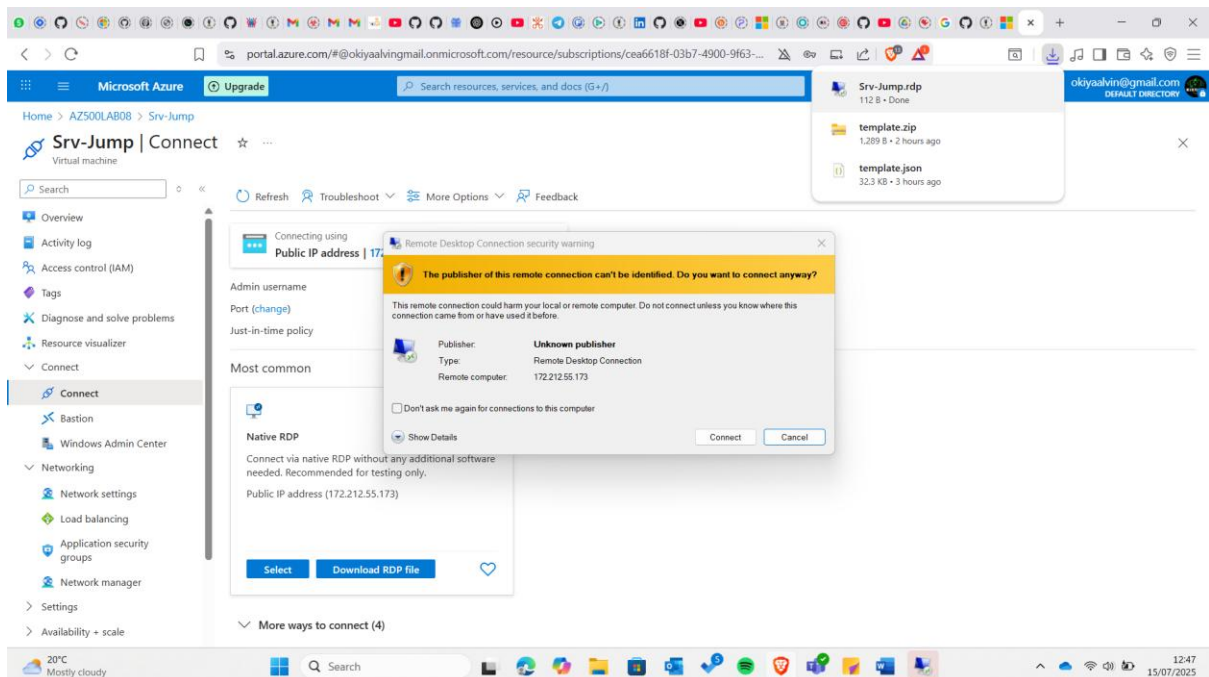
In this step I started by navigating to the the **AZ500LAB08** resource group and on the blade, in the list of resources,I clicked the **Srv-Jump** virtual machine.

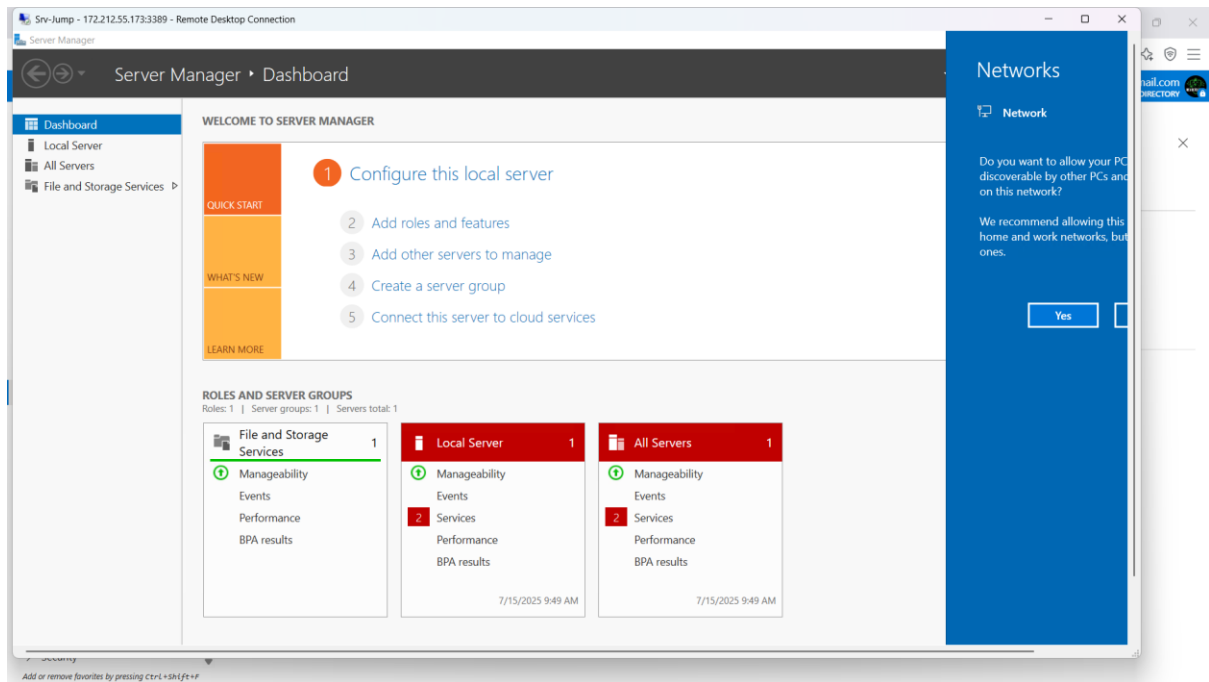


On the **Srv-Jump** blade, I clicked **Connect** and, in the drop down menu, clicked **Connect**,



Then I downloaded the RDP file and used it to connect to the **Srv-Jump** Azure VM via Remote Desktop. When prompted to authenticate, provide the username local admin and the password I used during the deployment of the custom template

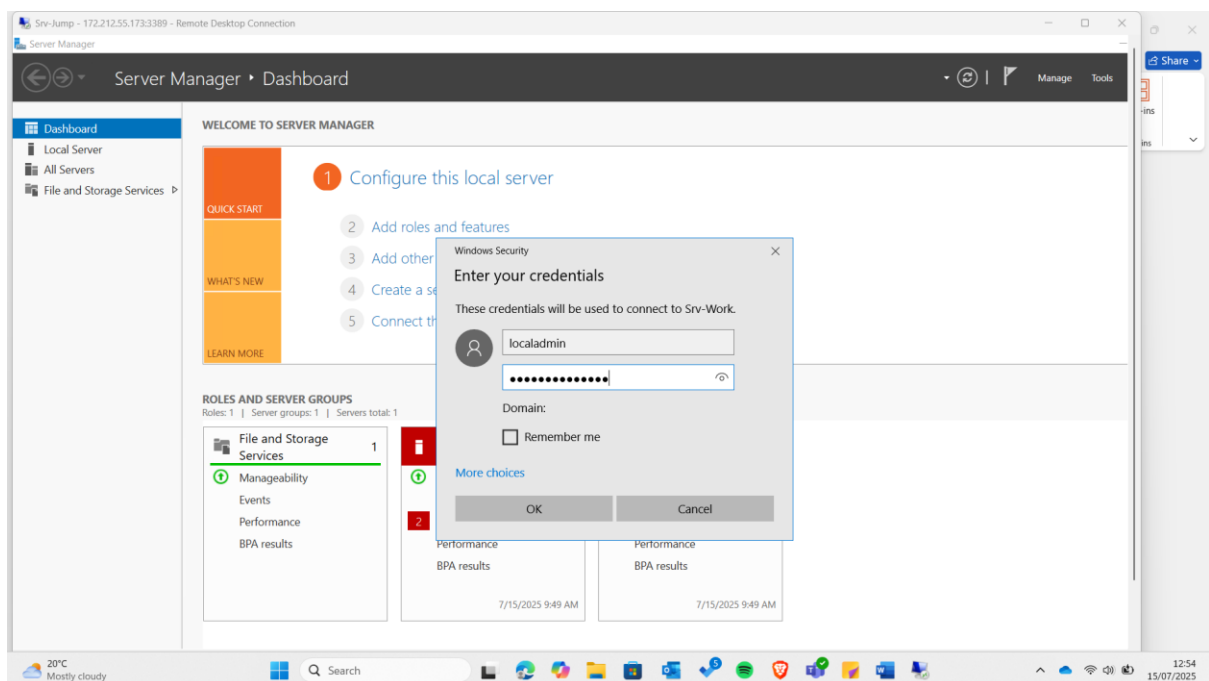




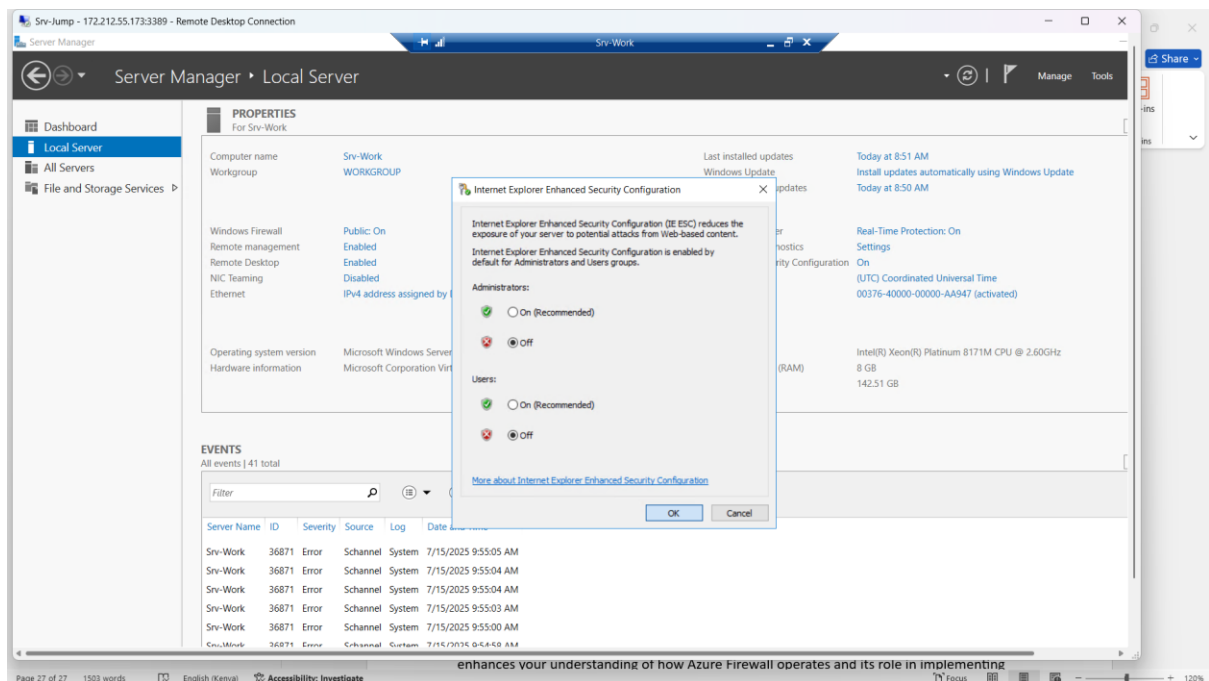
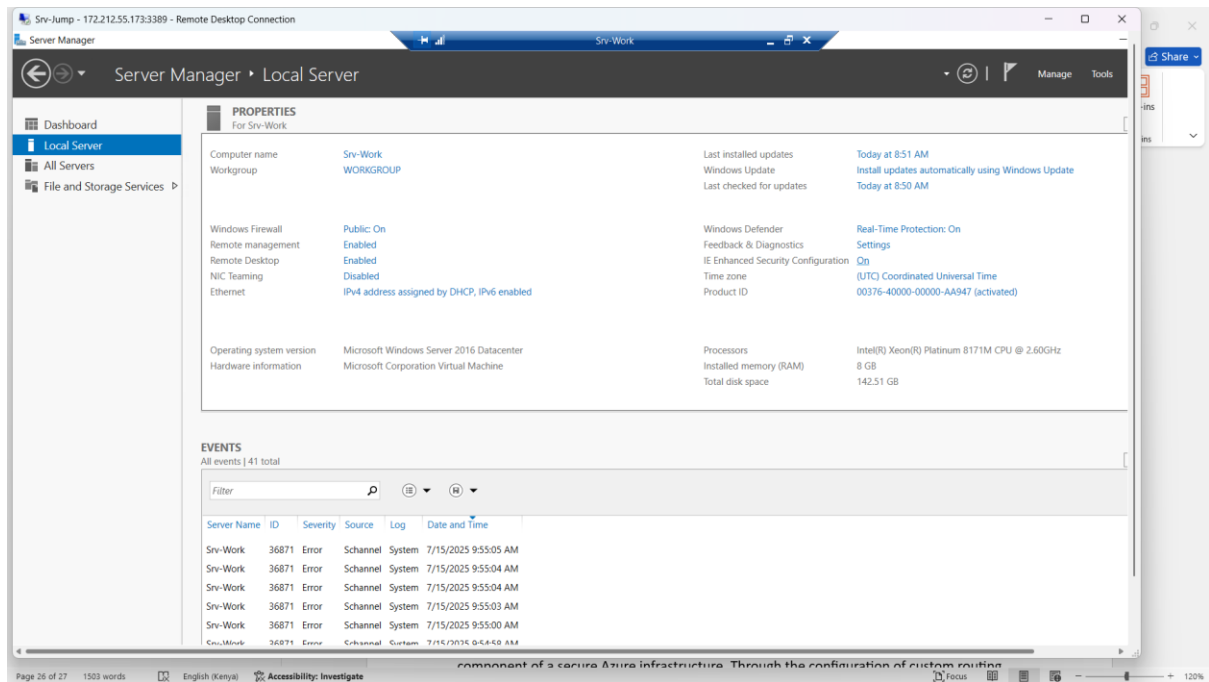
Within the Remote Desktop session to **Srv-Jump**, I right-clicked **Start**, in the right-click menu, clicked **Run**, and, from the **Run** dialog box, I ran the following to connect to **Srv-Work**:

mstsc /v:Srv-Work

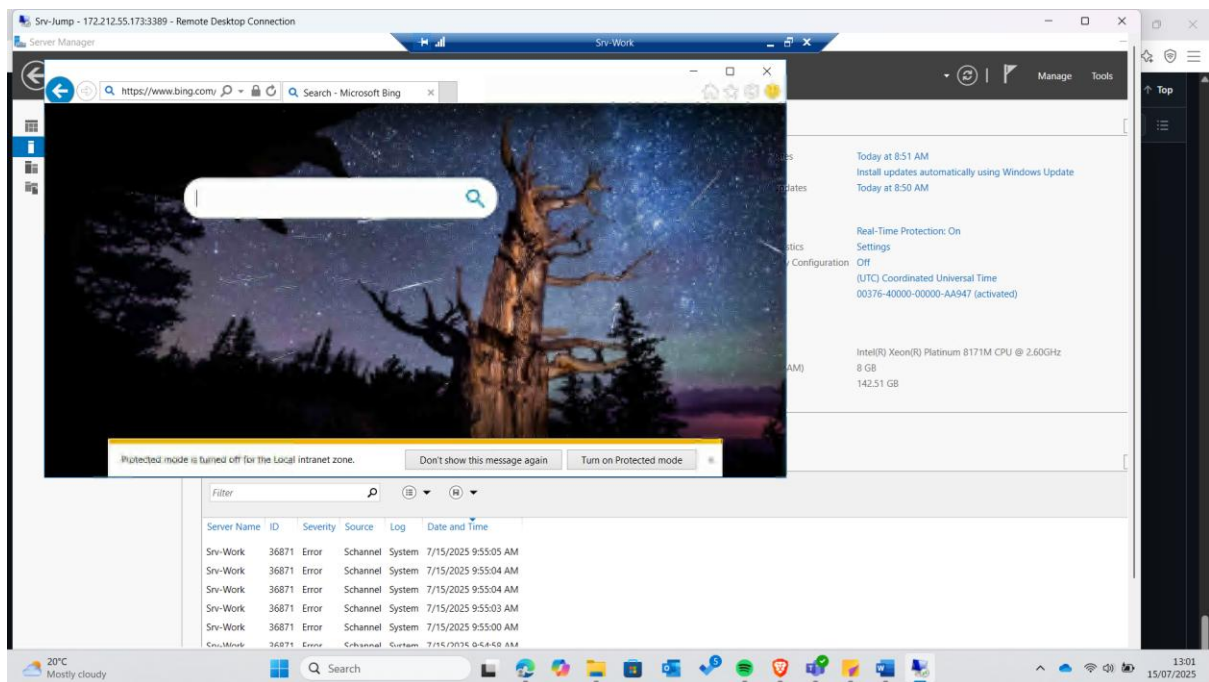
I used the same credentials to authenticate to srv work:



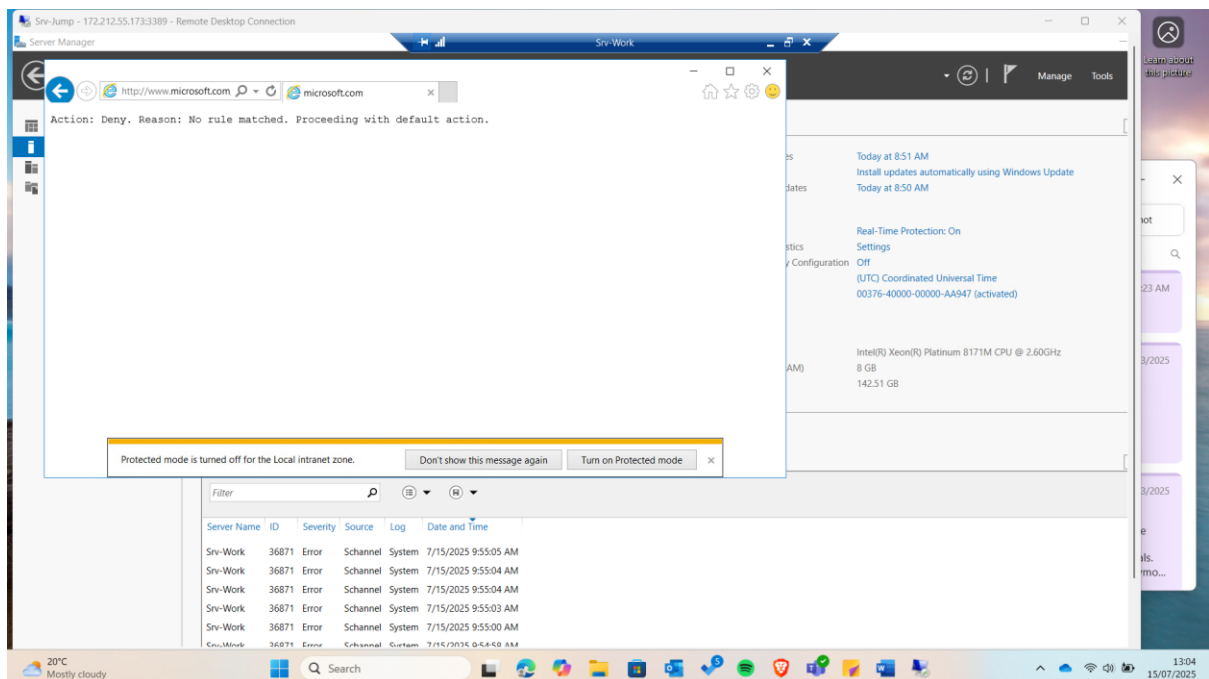
Within the Remote Desktop session to **Srv-Work**, in **Server Manager**, I clicked **Local Server** and then clicked **IE Enhanced Security Configuration** and disabled it by setting both options to **Off** and clicking **OK**



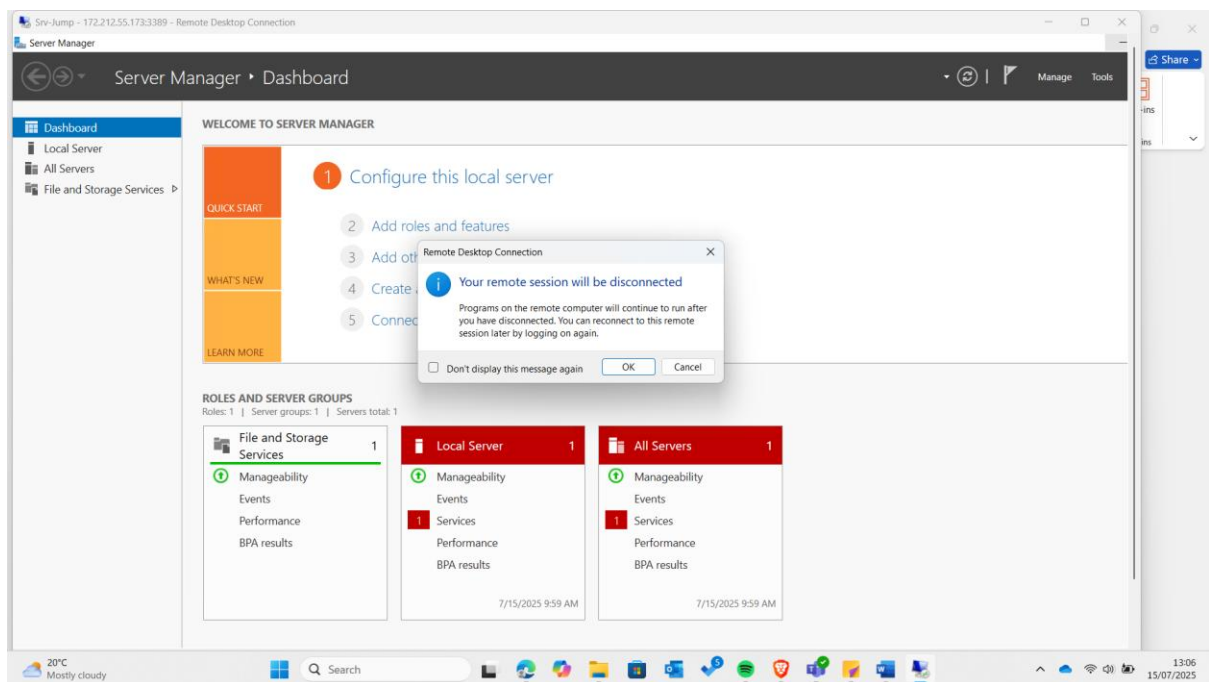
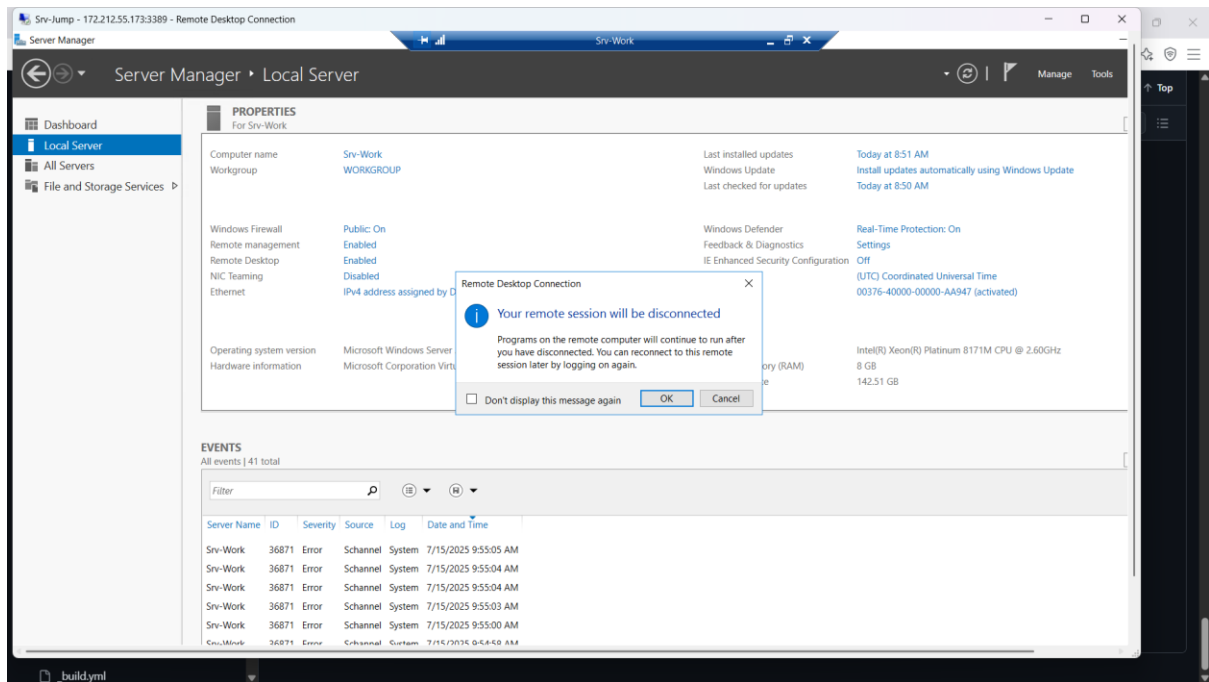
Attempting to load <https://www.bing.com> on browser loads successfully:



However trying another site like <http://www.microsoft.com/> brings an error:

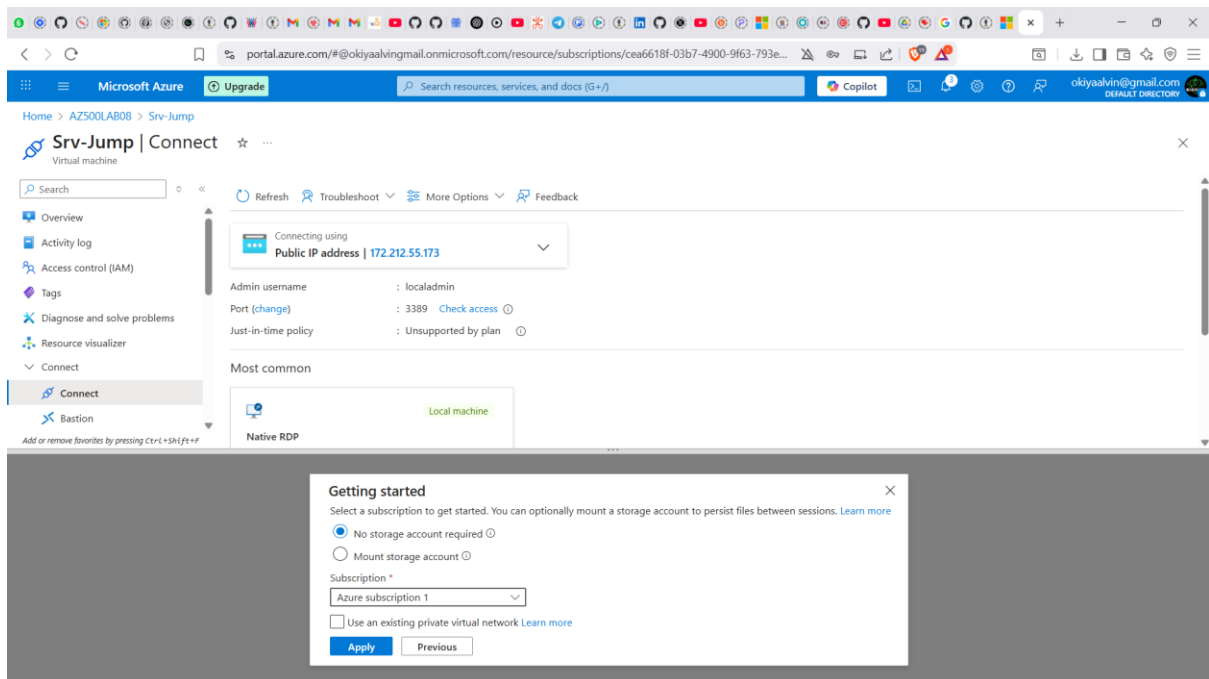
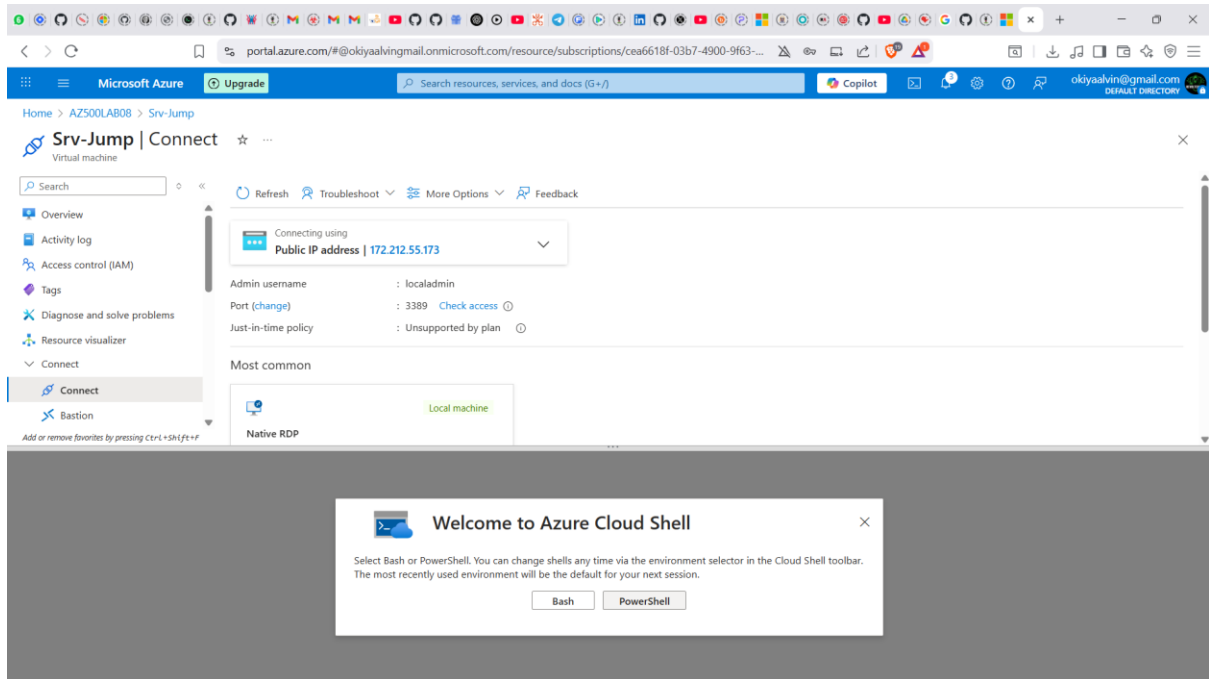


Having done tested the firewall I terminated the rdp sessions



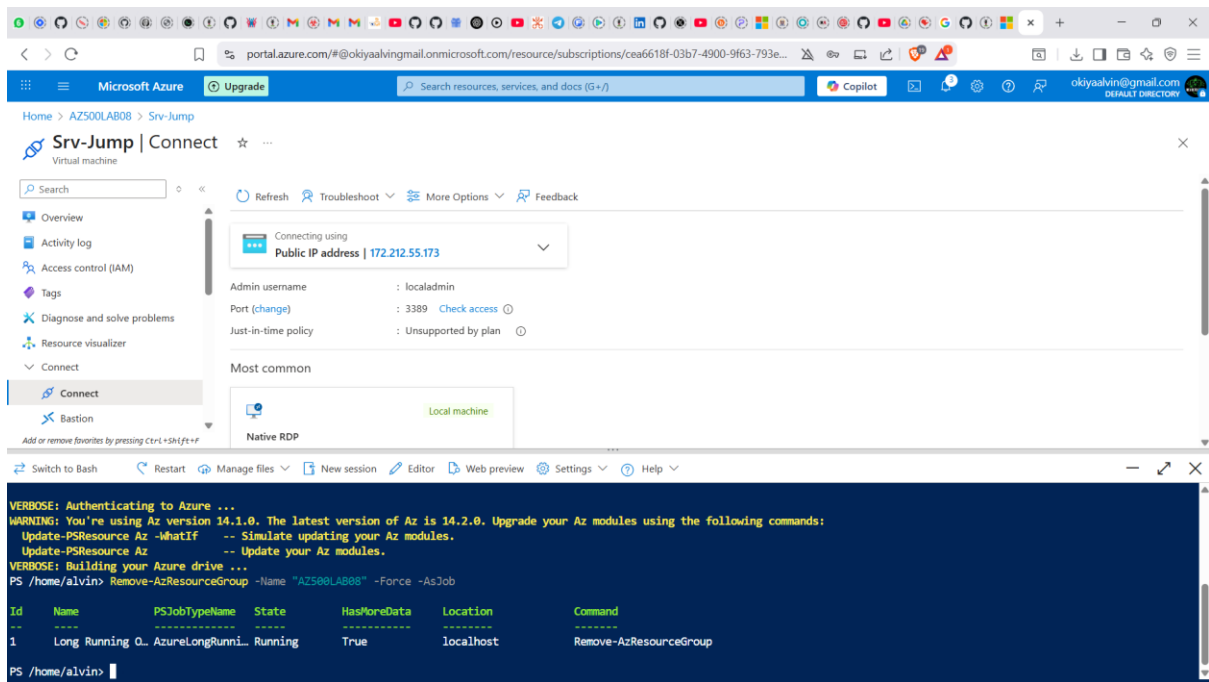
Clean up resources

1. I then cleaned up the resources by opening the Cloud Shell in the azure portal by clicking the first icon in the top right of the Azure Portal. when prompted, clicked **PowerShell** and **Create storage**.



I then ran the following command to clean up resources:

```
Remove-AzResourceGroup -Name "AZ500LAB08" -Force -AsJob
```



Conclusion

By completing this lab, you have successfully deployed and tested Azure Firewall as a critical component of a secure Azure infrastructure. Through the configuration of custom routing, application, and network rules, you enforced strict control over outbound traffic and demonstrated the firewall's ability to block unauthorized access. This hands-on experience enhances your understanding of how Azure Firewall operates and its role in implementing enterprise-grade network security within cloud environments. Always remember to clean up unused resources to avoid unnecessary charges and maintain an efficient cloud setup.